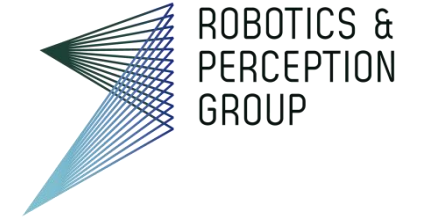




**University of
Zurich** ^{UZH}



Vision Algorithms for Mobile Robotics

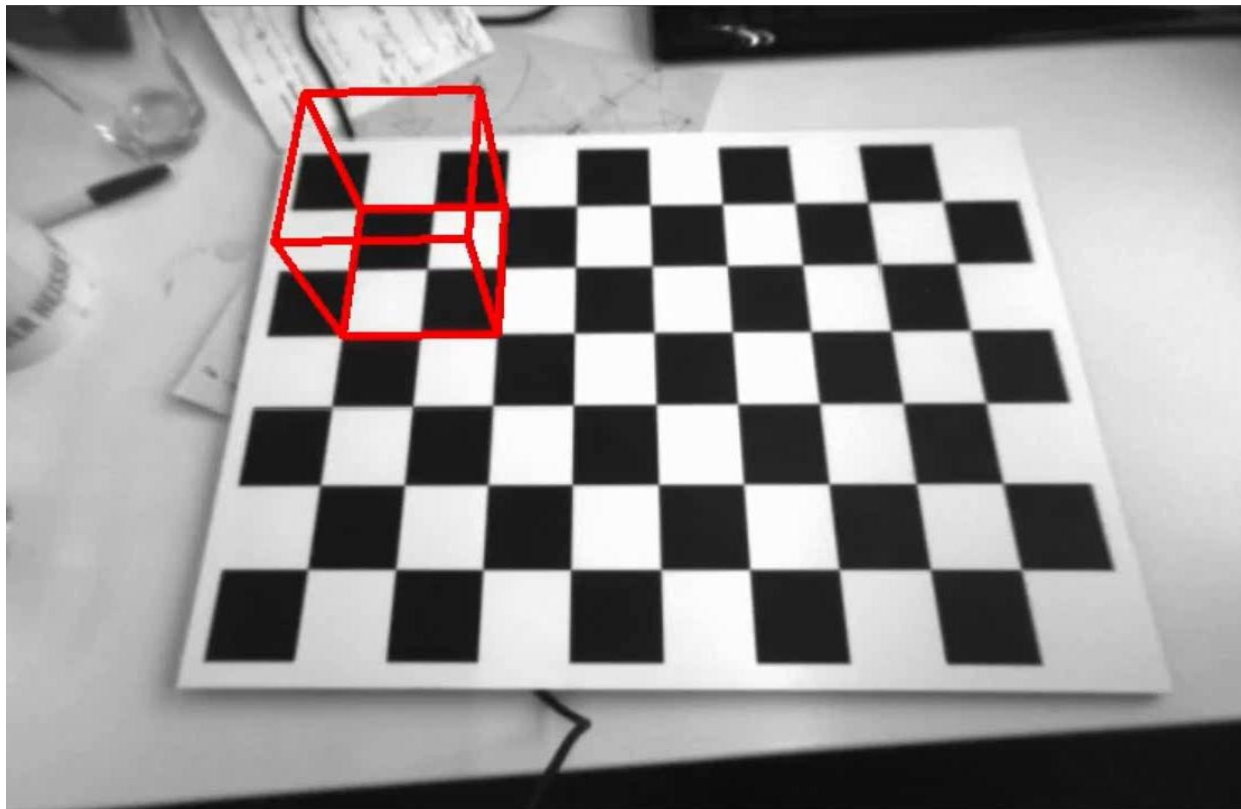
Lecture 02 Image Formation

Davide Scaramuzza

<https://rpg.ifi.uzh.ch>

Lab Exercise 1 - Today

Implement an augmented reality wireframe cube and practice the perspective projection

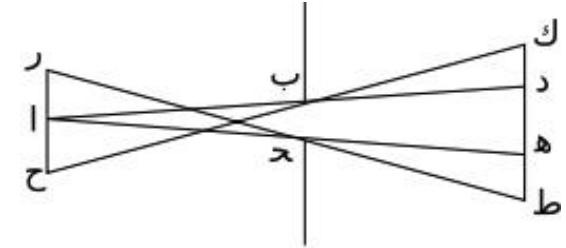


Today's Outline

- Image Formation
- Other camera parameters
- Digital camera
- Perspective camera model
- Lens distortion

Historical context

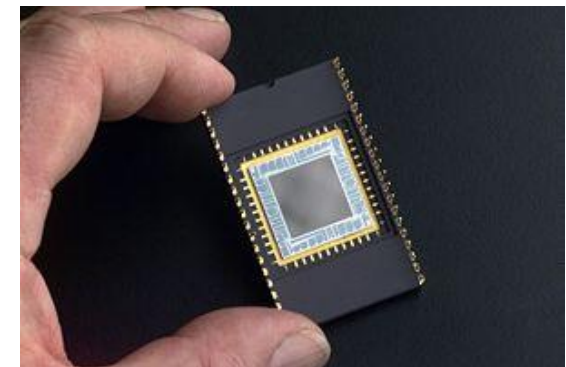
- **Pinhole model:** Mozi (470-390 BCE), Aristotle (384-322 BCE)
- Principles of optics (including lenses):
Alhacen (965-1039)
- **Camera obscura:** first scientific description by Arabian scientist Ibn al-Haytham (965-1038), who inspired Leonardo da Vinci (1452-1519) and others
- **First photo (heliography):** Joseph Nicephore Niepce (1827)
- Daguerréotypes (1839)
- **Photographic film** (Eastman, **1888**, **founder of Kodak**)
- Cinema (Lumière Brothers, 1895)
- Color Photography (Lumière Brothers, 1908)
- Television (Baird, Farnsworth, Zworykin, 1920s)
- **First consumer camera with CCD:** Sony Mavica (1981) (output was analog and stored on magnetic disk)
- **First fully digital camera:** Kodak DCS100 (1990)



Alhacen's notes



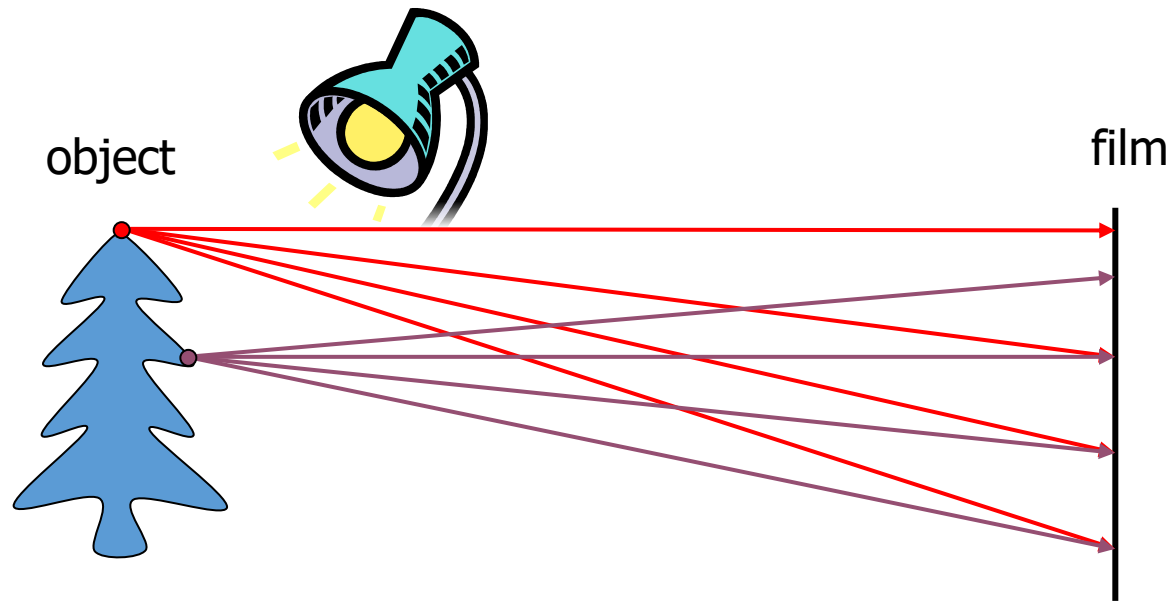
First surviving photo ever: "View from the Window at Le Gras", by Niepce, 1827



CCD chip

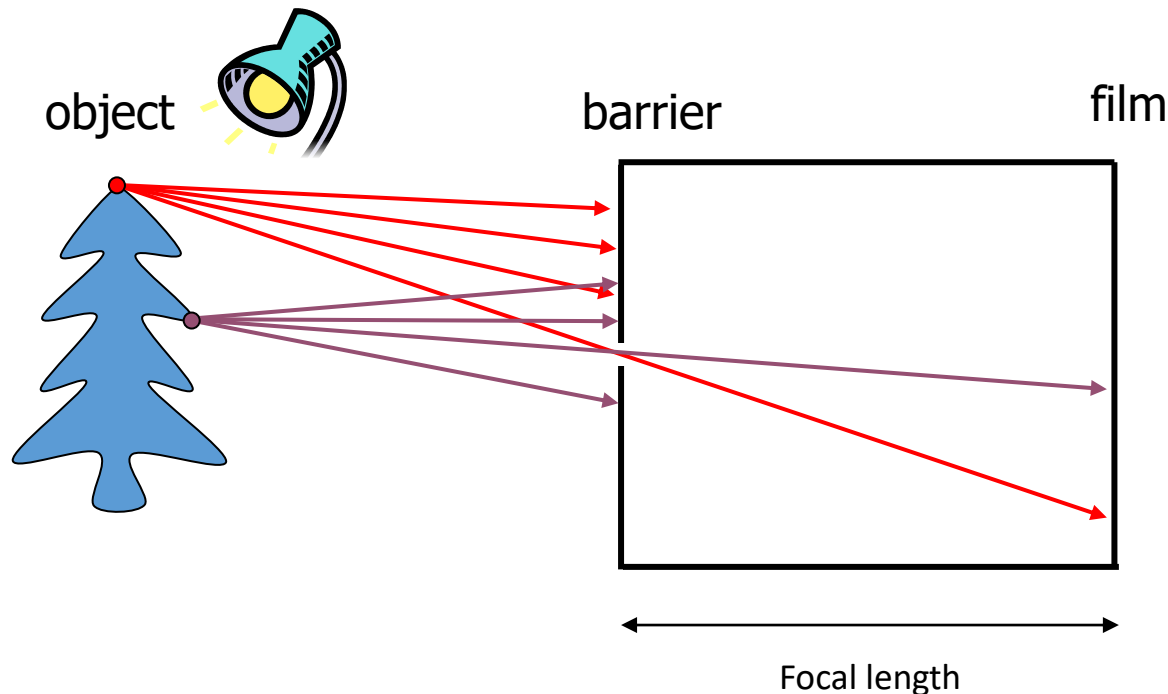
How to form an image

- Would this work?



Camera obscura

- Add a **barrier** with an **opening** to block off most of the rays
 - This reduces blurring, i.e., superimposition of multiple rays on any point of the film
 - The opening is called “**aperture**”
 - The **depth of the room** is called “**focal length**”



Home-made camera obscura

- The image is **inverted**
- How can we **reduce the blur**?

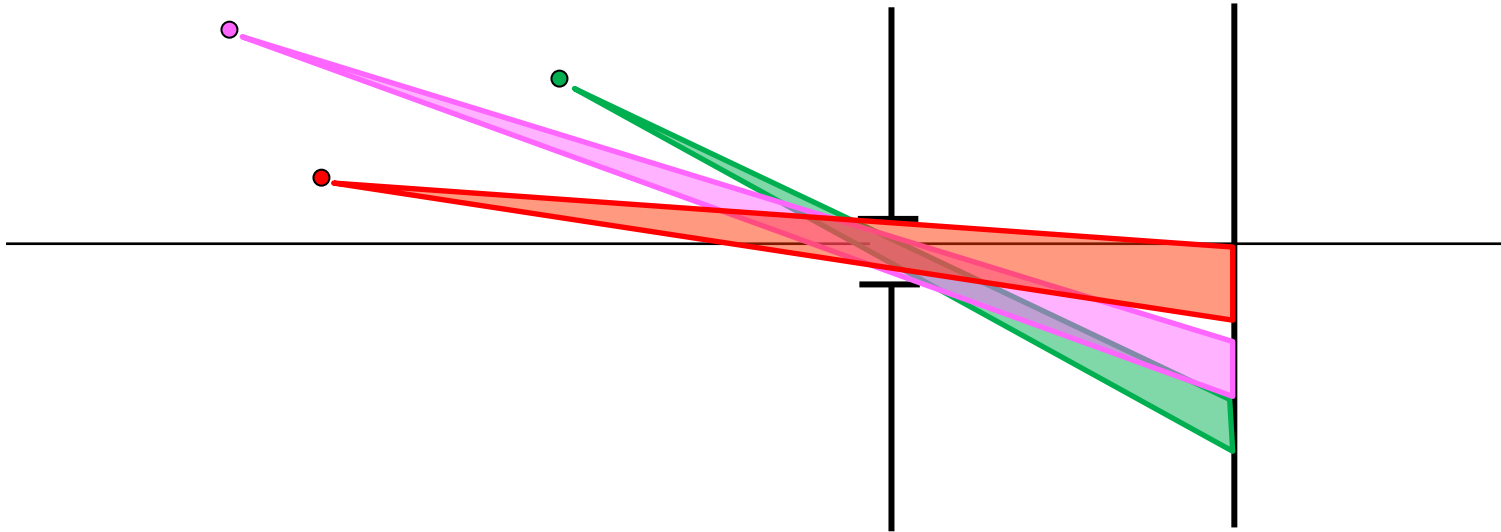


How to build a camera obscura at home ([link](#))



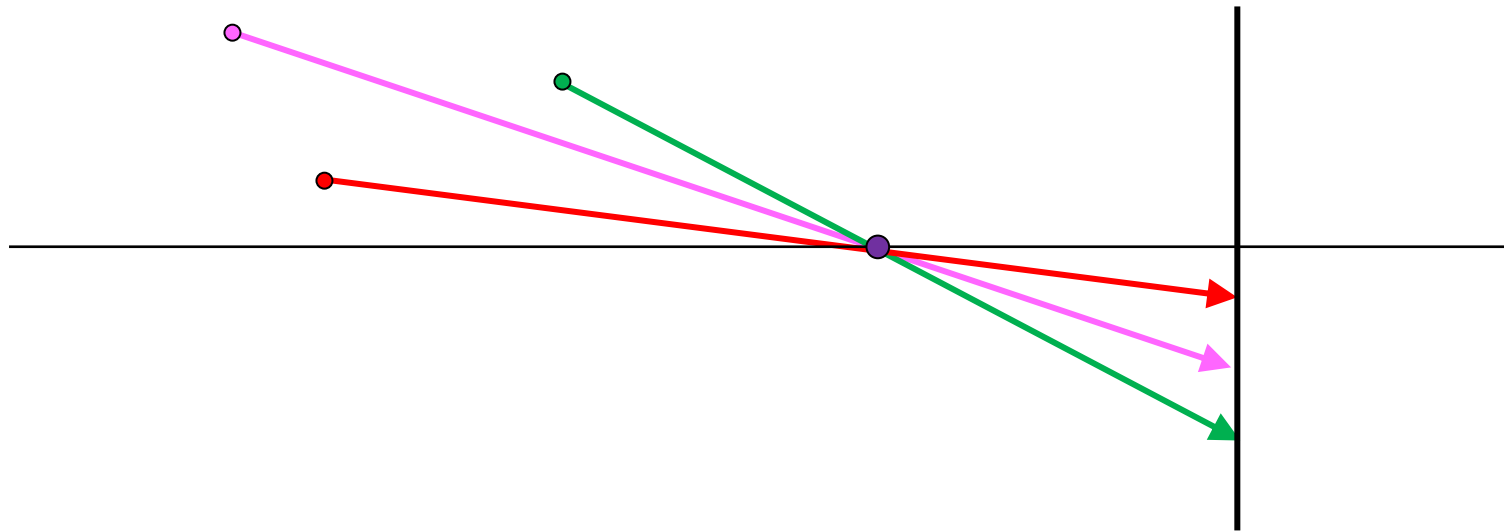
Effects of the Aperture Size

- A large **aperture** makes the **image blurry** because a cone of light is let through from each world point



Effects of the Aperture Size

- **Shrinking the aperture** makes the **image sharper**
- The ideal aperture is a **pinhole** that only lets through one ray of light from each world point



Why not making the aperture as small as possible?

- With **small apertures**, less light gets **through** → must **increase exposure time**
- If aperture gets **too small**, **diffraction effects** start to appear

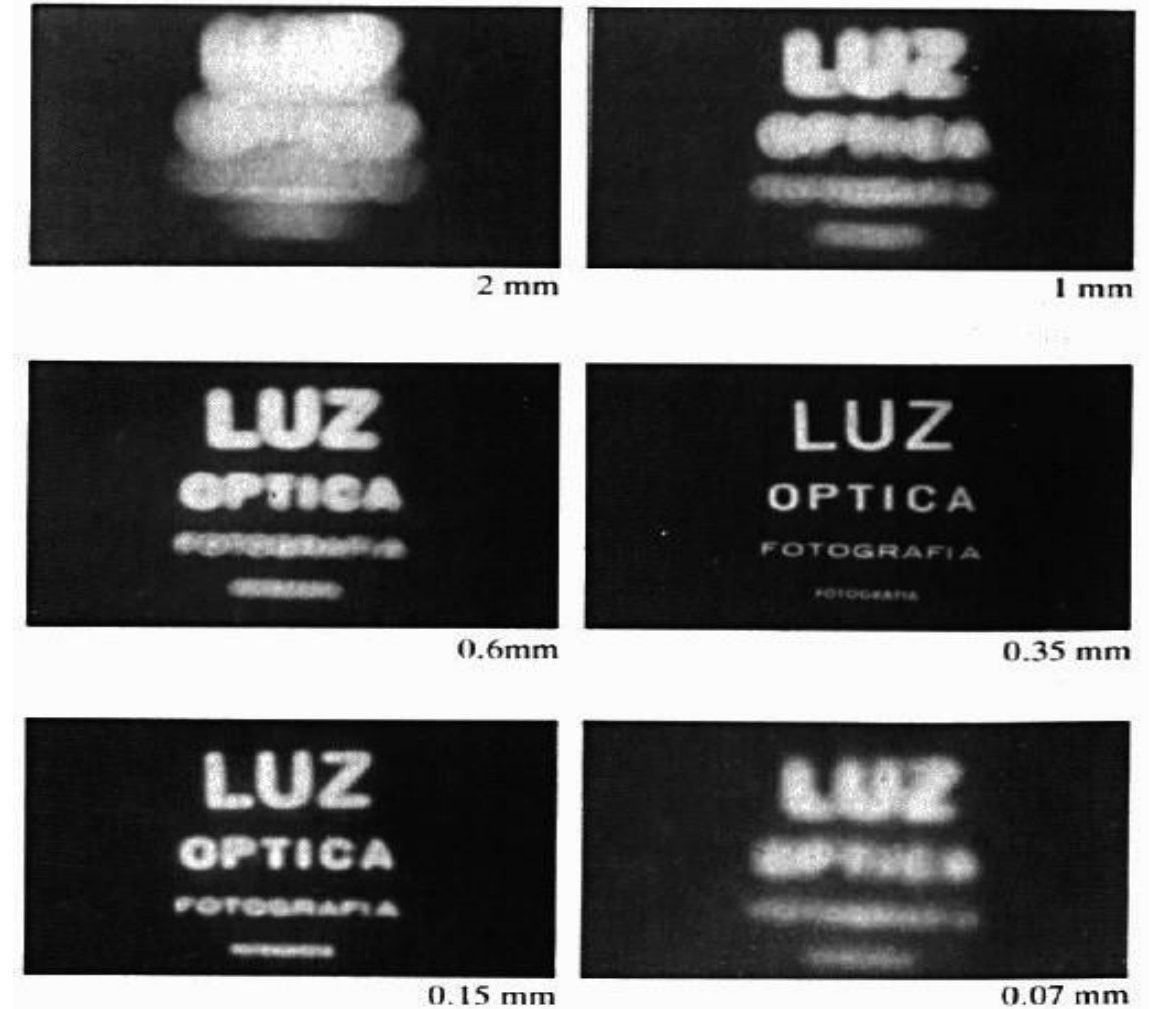


Image formation using a converging lens

- A **thin converging lens** focuses light onto the film satisfying two properties:

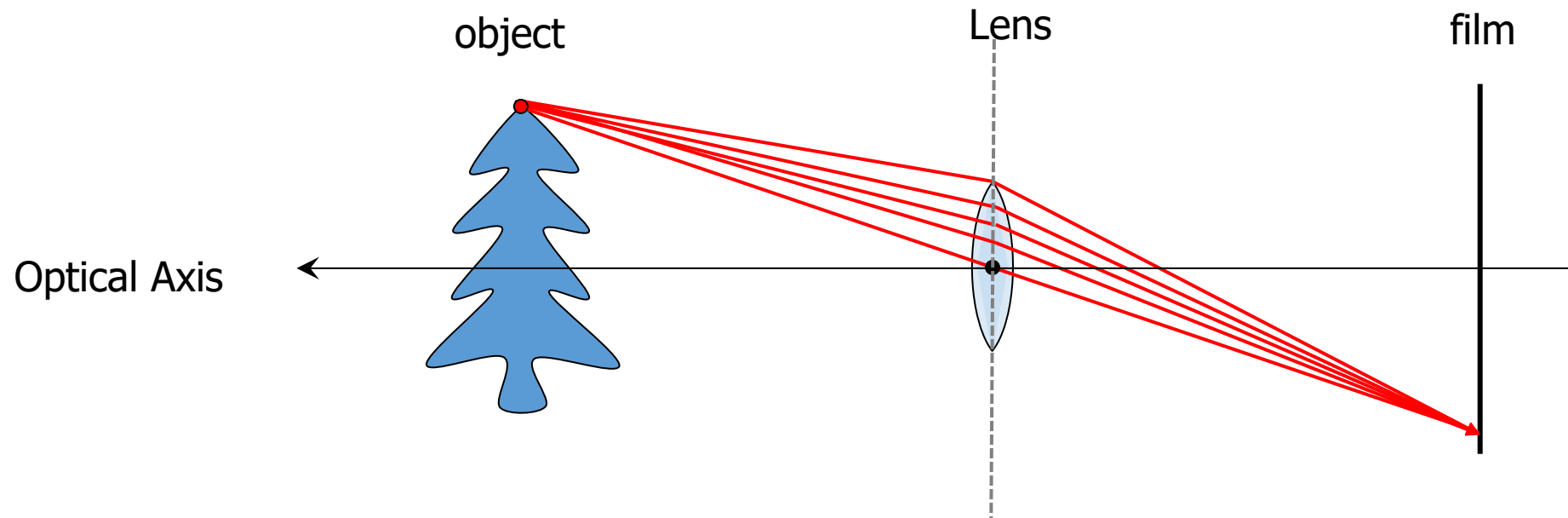


Image formation using a converging lens

- A **thin converging lens** focuses light onto the film, satisfying **two properties**:
 1. Rays passing through the **Optical Center** are not deviated

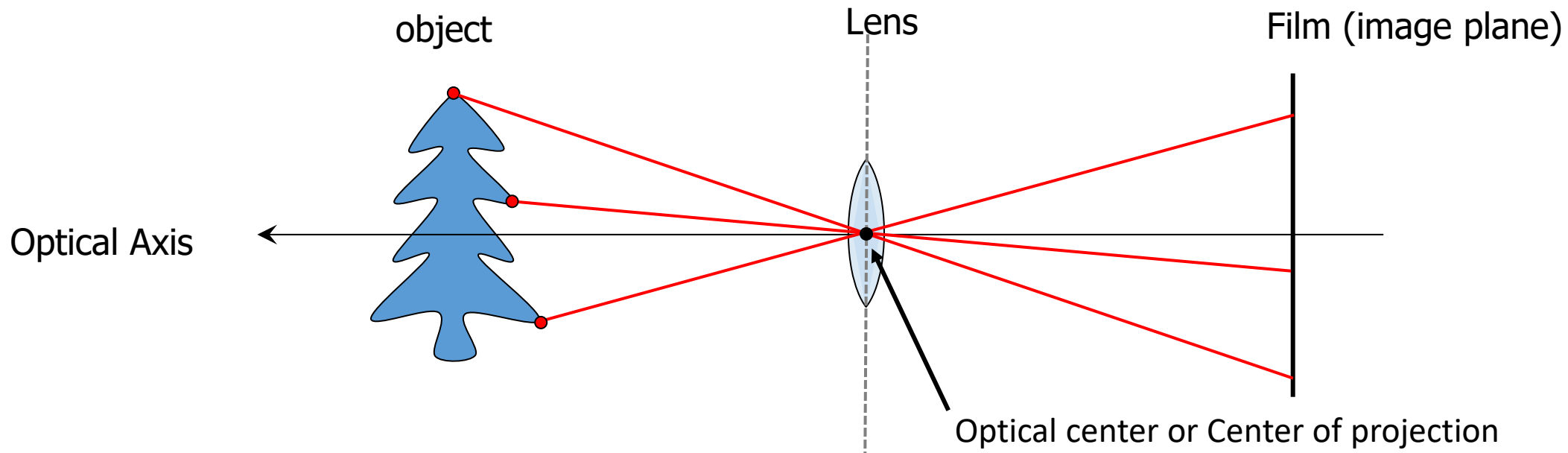
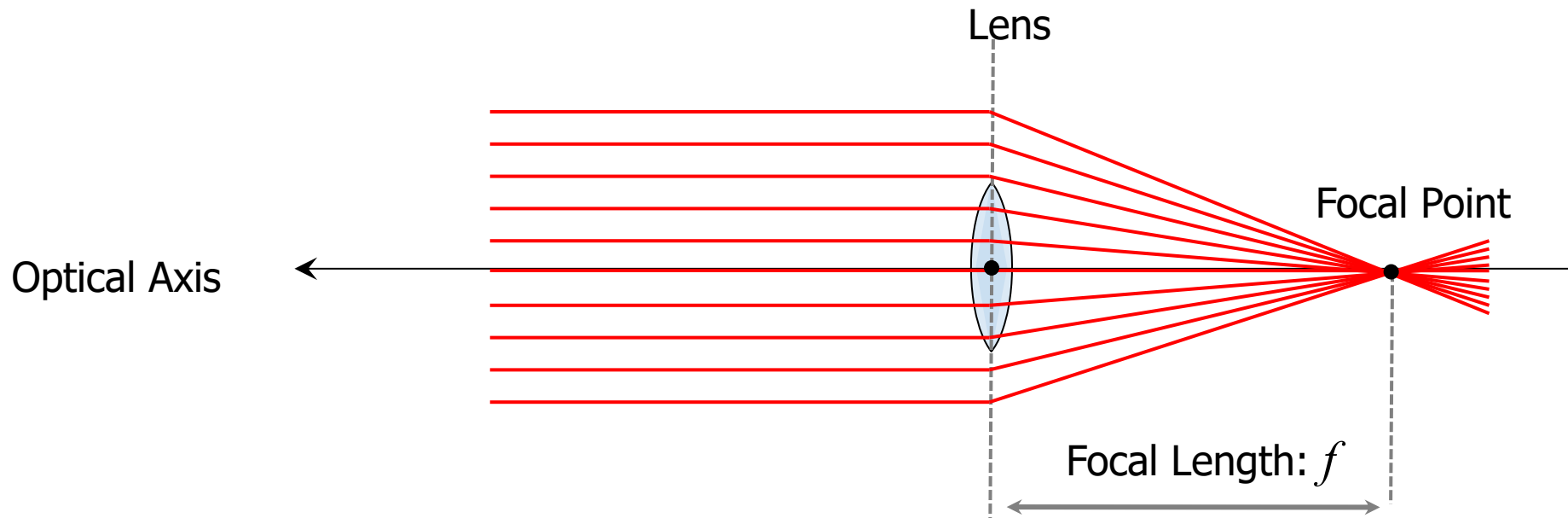


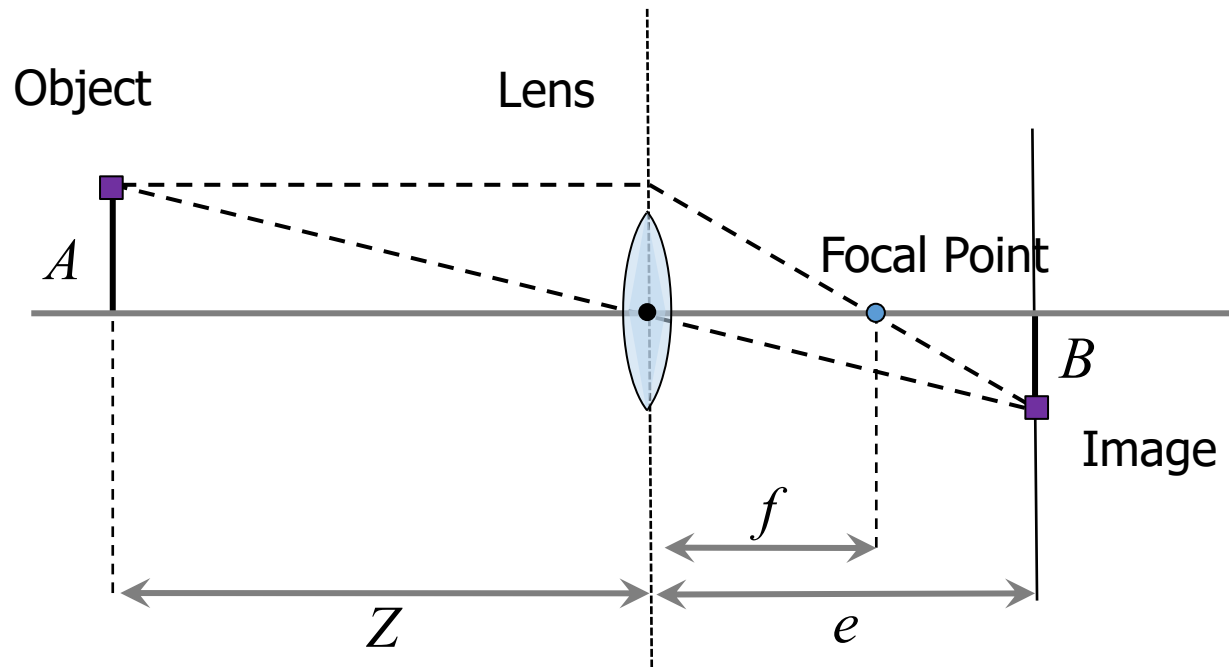
Image formation using a converging lens

- A **thin converging lens** focuses light onto the film satisfying two properties:
 1. Rays passing through the **Optical Center** are not deviated
 2. All rays parallel to the **Optical Axis** converge at the **Focal Point**



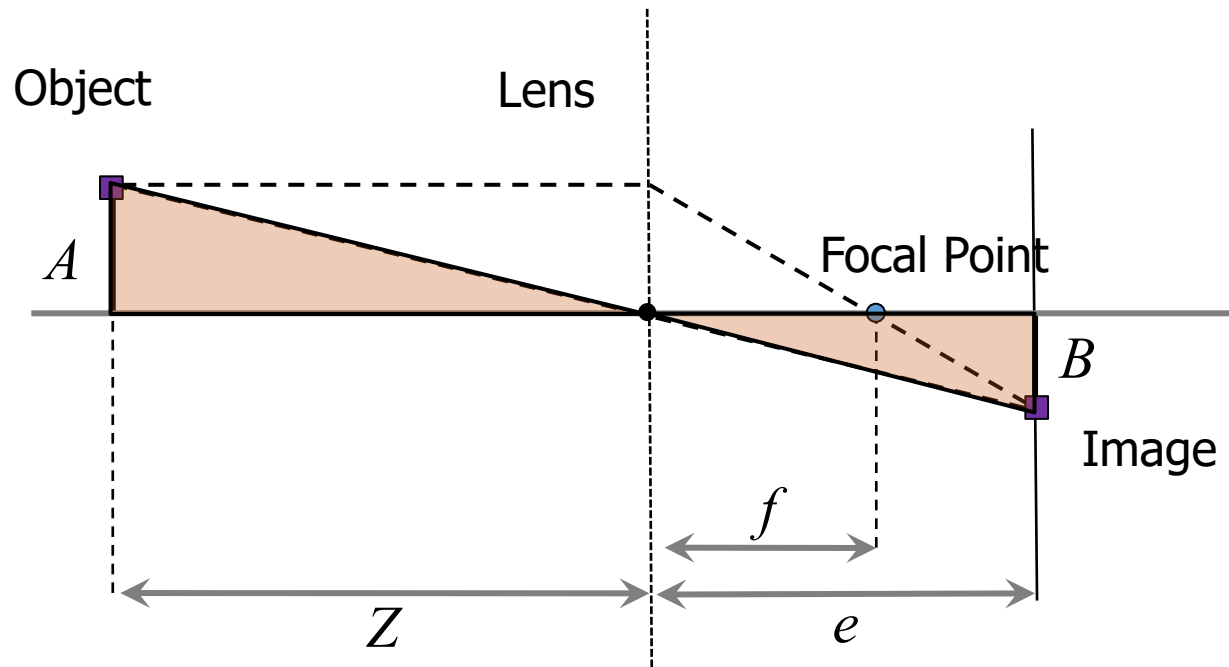
Thin lens equation

- Find a relationship between f , Z , and e



Thin lens equation

- Find a relationship between f , Z , and e

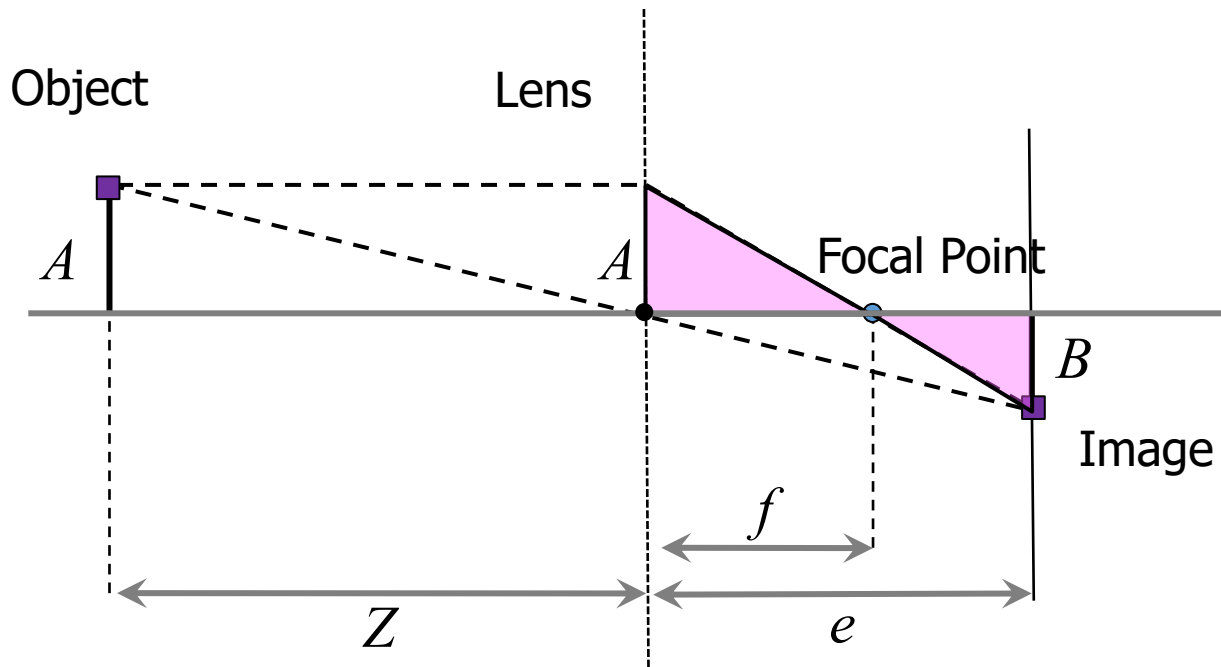


Similar Triangles:

$$\frac{B}{A} = \frac{e}{Z}$$

Thin lens equation

- Find a relationship between f , Z , and e



Similar Triangles:

$$\frac{B}{A} = \frac{e}{Z}$$

$$\frac{B}{A} = \frac{e-f}{f} = \frac{e}{f} - 1$$

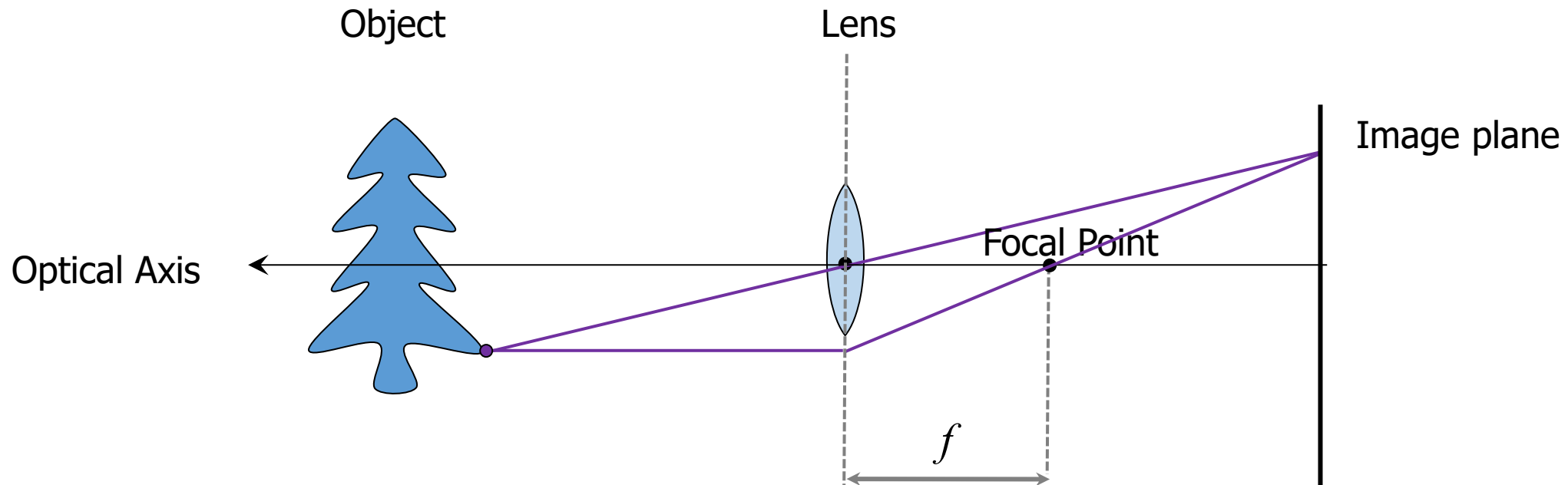
$$\left. \begin{array}{l} \frac{B}{A} = \frac{e}{Z} \\ \frac{B}{A} = \frac{e-f}{f} = \frac{e}{f} - 1 \end{array} \right\} \frac{e}{f} - 1 = \frac{e}{Z} \Rightarrow \boxed{\frac{1}{f} = \frac{1}{Z} + \frac{1}{e}}$$

"Thin lens equation"

- Any object point satisfying this equation is in focus
- Can I use this to measure distances?

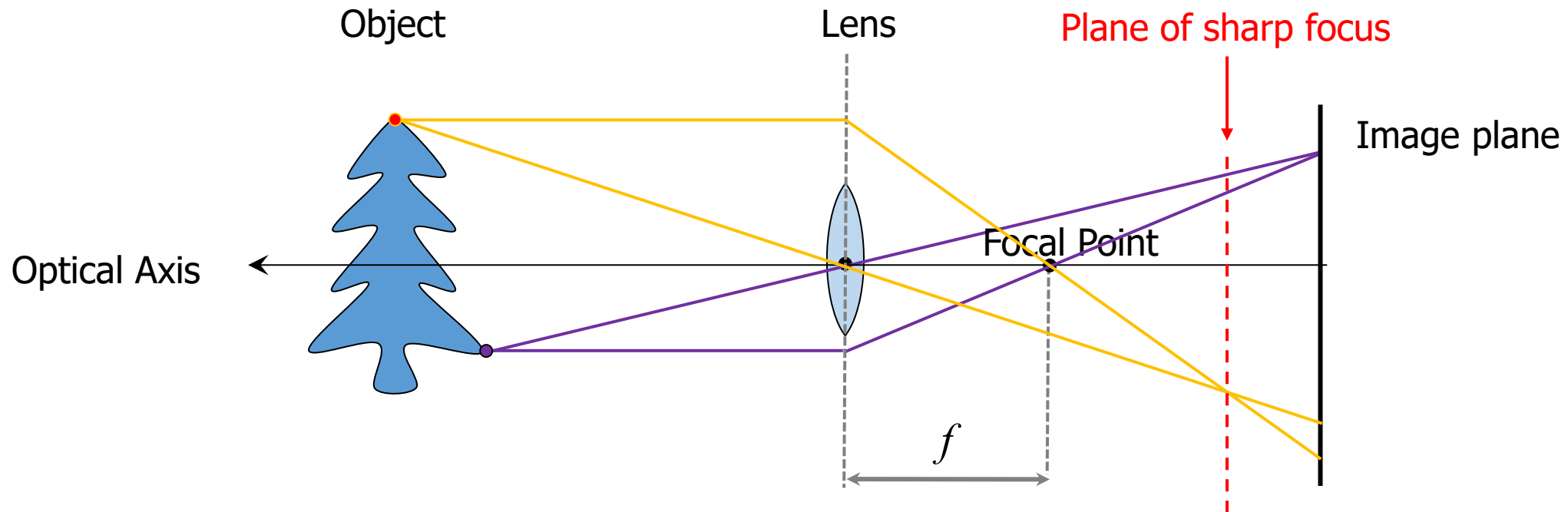
“In focus”

- For a given point on the object, there is a specific distance between the lens and the film, at which the object appears **in focus** in the image



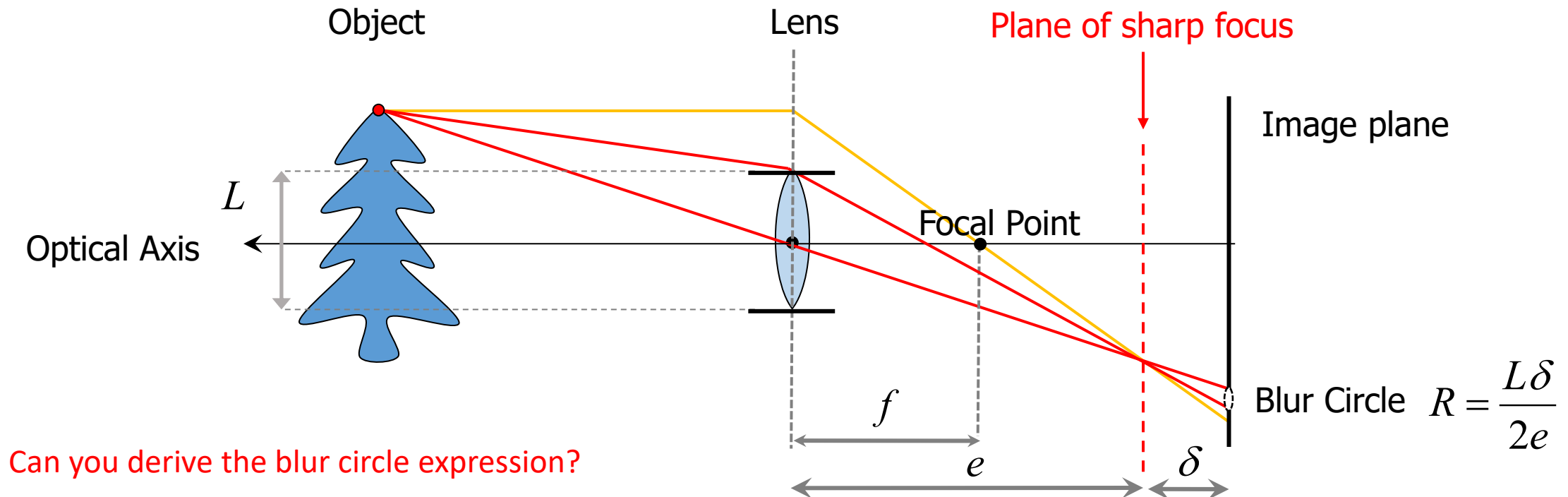
“In focus”

- For a given point on the object, there is a specific distance between the lens and the film, at which the object appears **in focus** in the image
- Other points project to a **blur circle**



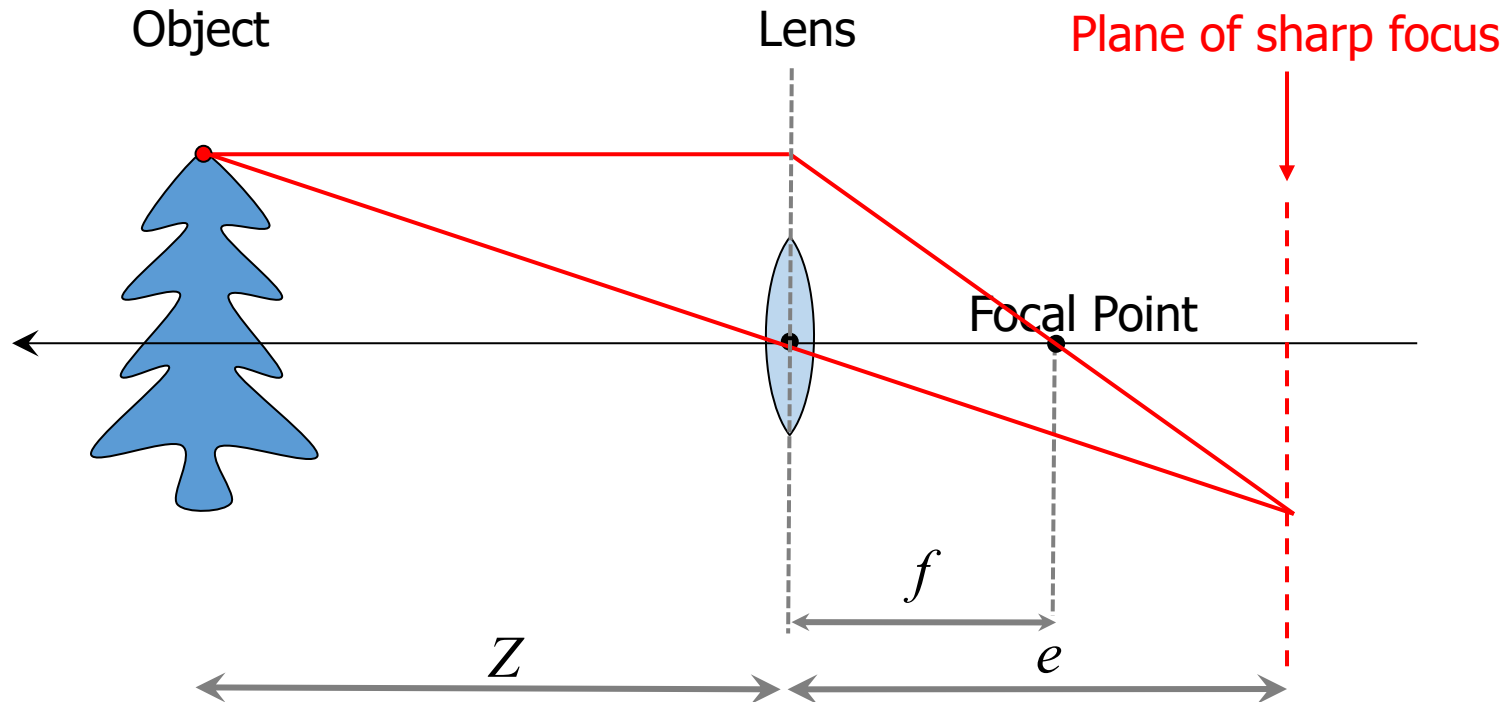
Blur Circle

- The blur circle has radius: $R = L\delta/(2e)$
 - Observe how both **small aperture (L)** and δ yield a **small blur circle (R)**
 - To capture a sharp image, we must **adjust the camera settings** such that $2R$ remains **smaller than the 1 pixel**



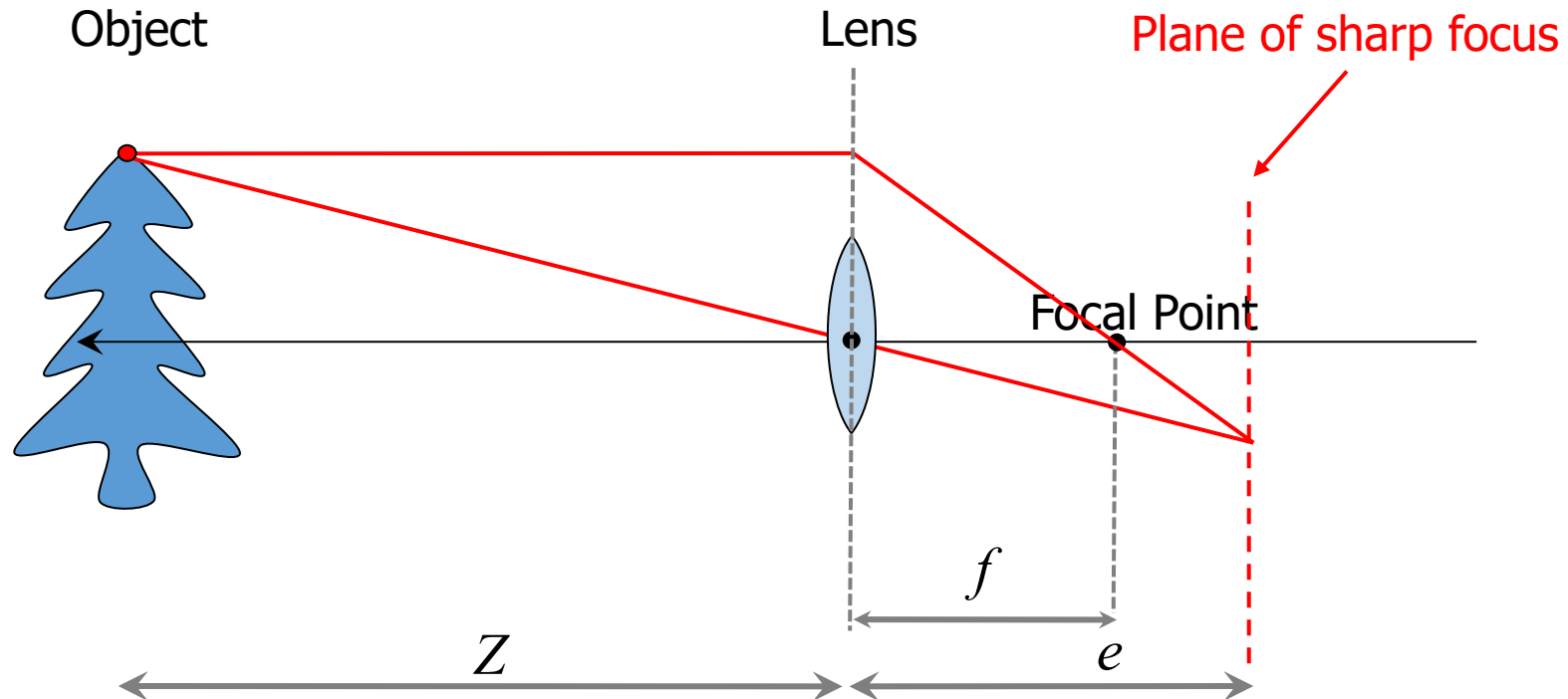
The Pin-hole approximation

- What happens if $z \gg f$?



The Pin-hole approximation

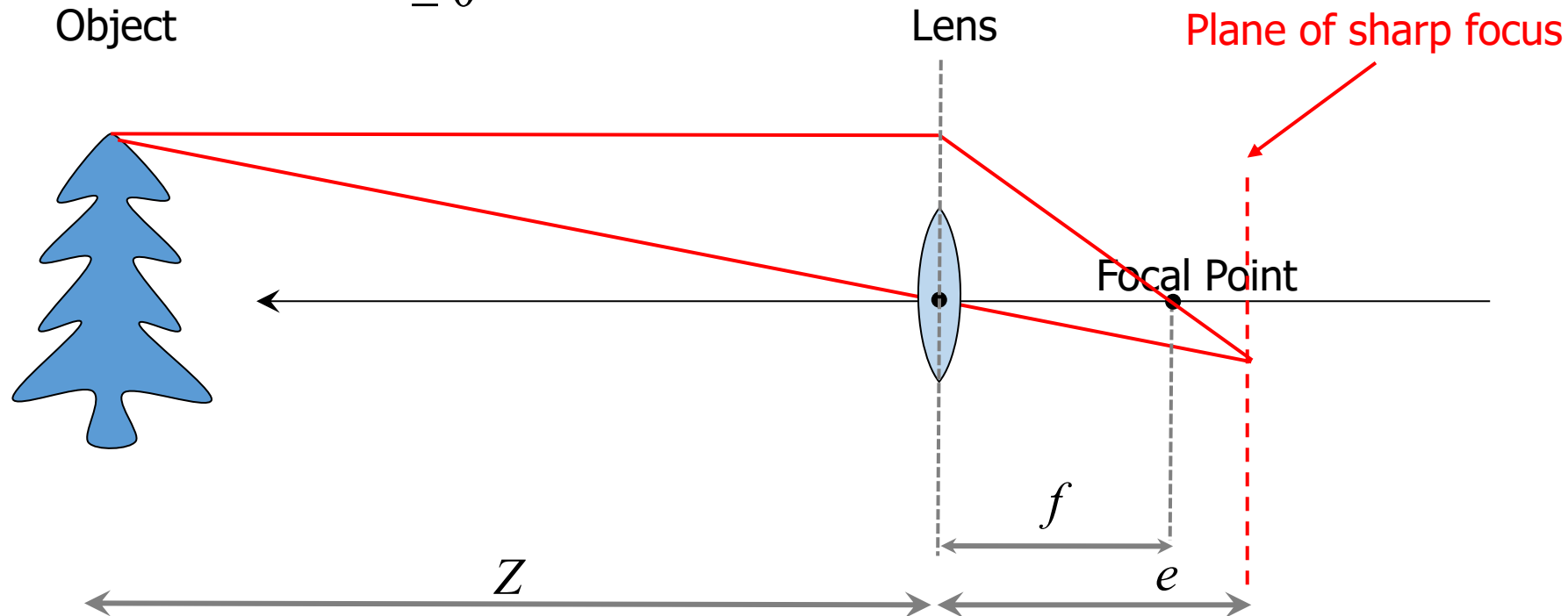
- What happens if $z \gg f$?



The Pin-hole approximation

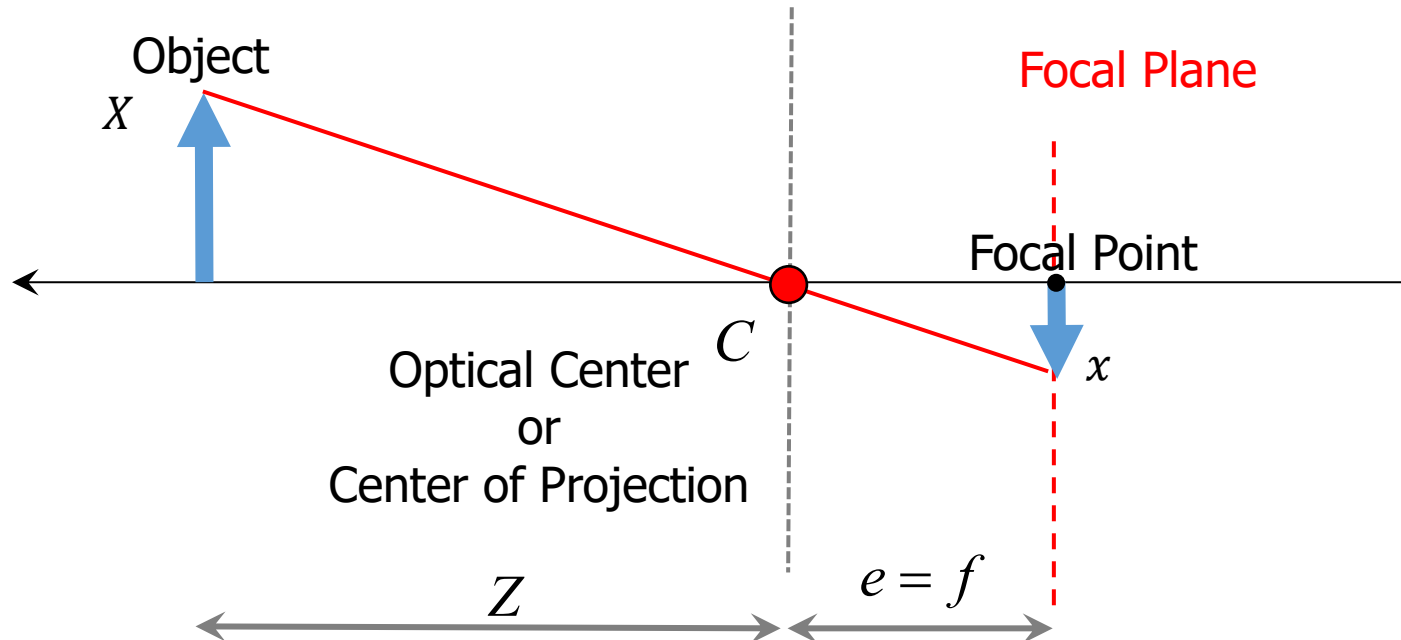
- What happens if $z \gg f$?

- We observe that $\frac{1}{f} = \frac{1}{z} + \frac{1}{e} \Rightarrow \frac{1}{f} \approx \frac{1}{e} \Rightarrow f \approx e \Rightarrow$ focal plane approaches the focal point



The Pin-hole approximation

- The **Pinhole Approximation** allows us to replace the bundle of rays from each object point with the single ray through the lens center.
- The relation between the image and object becomes: $-\frac{x}{X} = \frac{f}{Z} \Rightarrow x = -f \frac{X}{Z}$
- This is called **Perspective Projection**



Perspective effects

Far away objects appear smaller, with **size inversely proportional to distance**



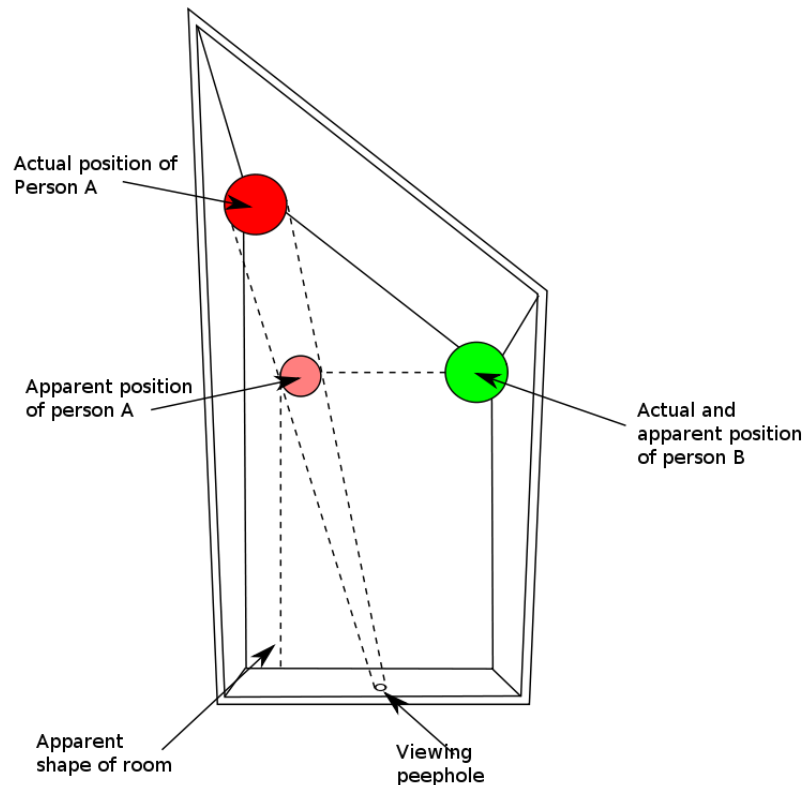
Perspective can be used as a prior

Perspective gives us very strong depth cues and, thanks to experience (e.g., Manhattan world assumption), we **can perceive a 3D scene by viewing its 2D representation** (i.e. the image)



Playing with Perspective: the Ames room

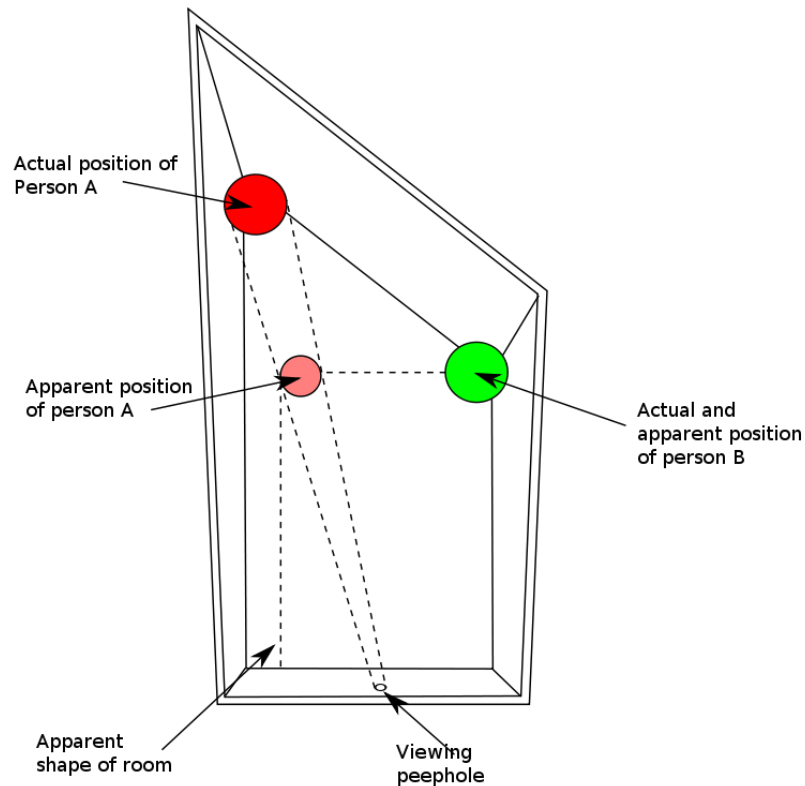
However, our perception of a 3D scene can be fooled. An example is the Ames' room (invented in 1946). Check out the **Ames room** in the **Technorama** science museum in Winterthur)



Ames room documentary by neuroscientist
Dr. V.S. Ramachandran:
<https://youtu.be/TtdOYjXF0no>

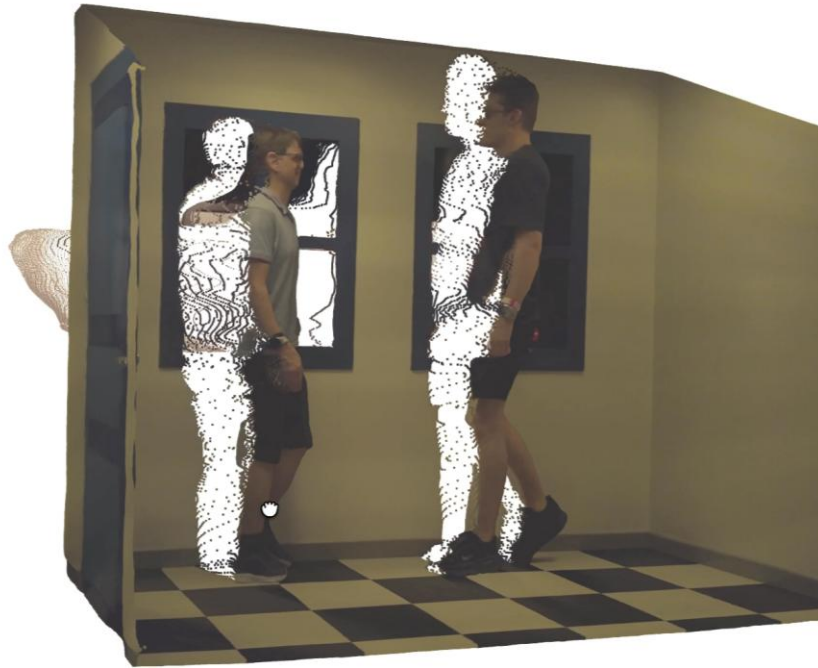
Playing with Perspective: the Ames room

However, our perception of a 3D scene can be fooled. An example is the Ames' room (invented in 1946). Check out the **Ames room** in the **Technorama** science museum in Winterthur)



The Ames Room Can Also Fool Learned Depth Estimation!

Predicted 3D reconstruction

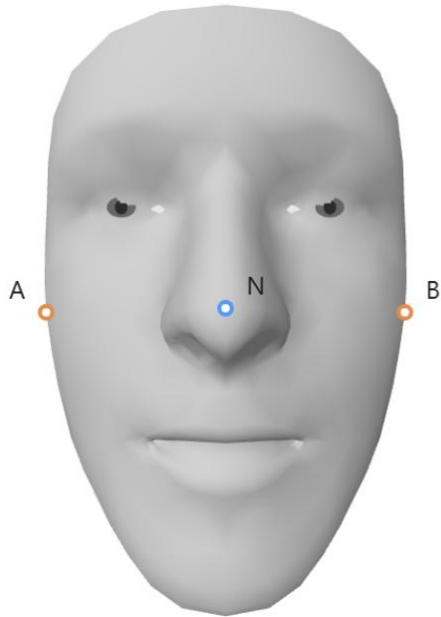


Predicted depth (DA3)

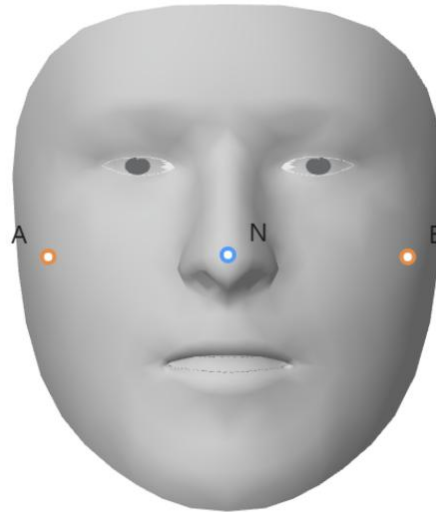


Why does your nose look bigger in a selfie than in real life?

Home exercise: Use the perspective projection equations to explain why, in a picture of a face taken frontally and from a very small distance, the nose appears much larger than the rest of the face. What's the distortion factor? Can this effect be reduced by acting on the focal length? **(This question was asked at the 2022 exam)**



Near-by selfie



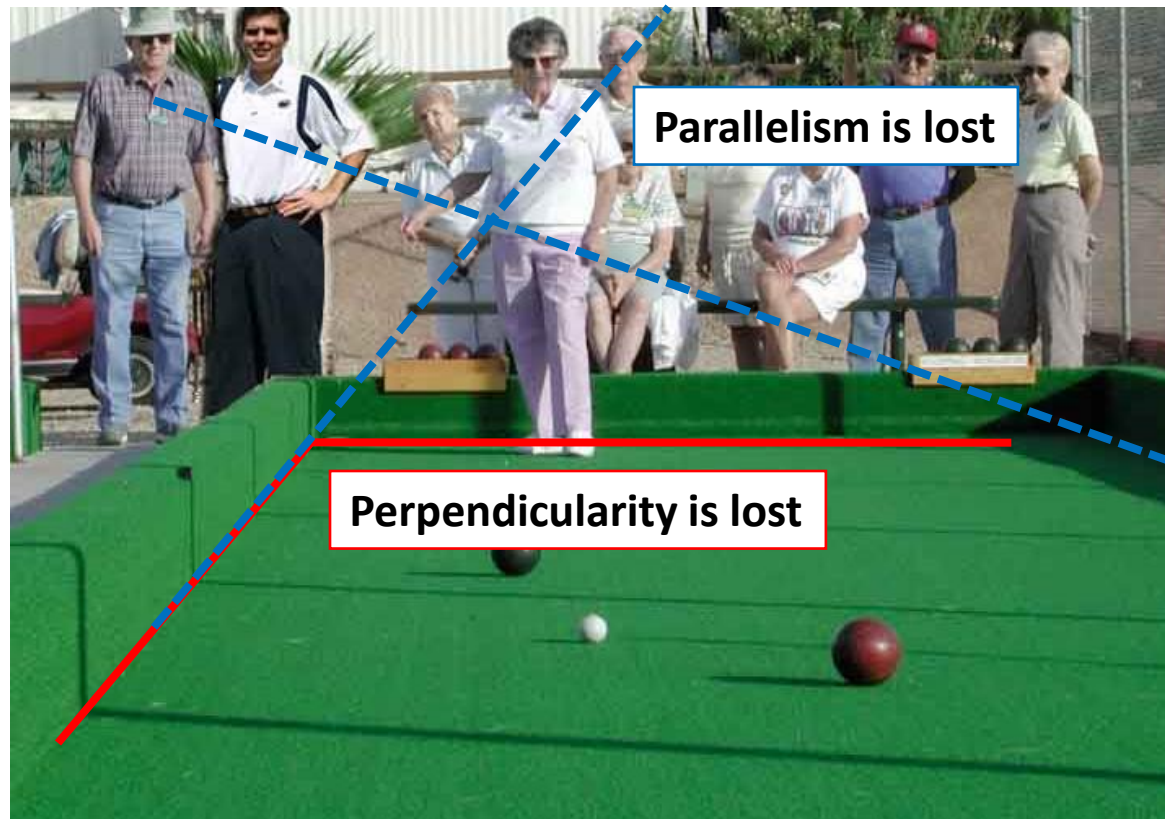
Far-distance photo



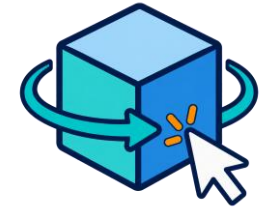
[Explore this phenomenon
with interactive
visualization](#)

Perspective Projection: what is preserved or lost?

- **Straight lines** are still **straight**
- **Lengths and angles** are **not preserved**

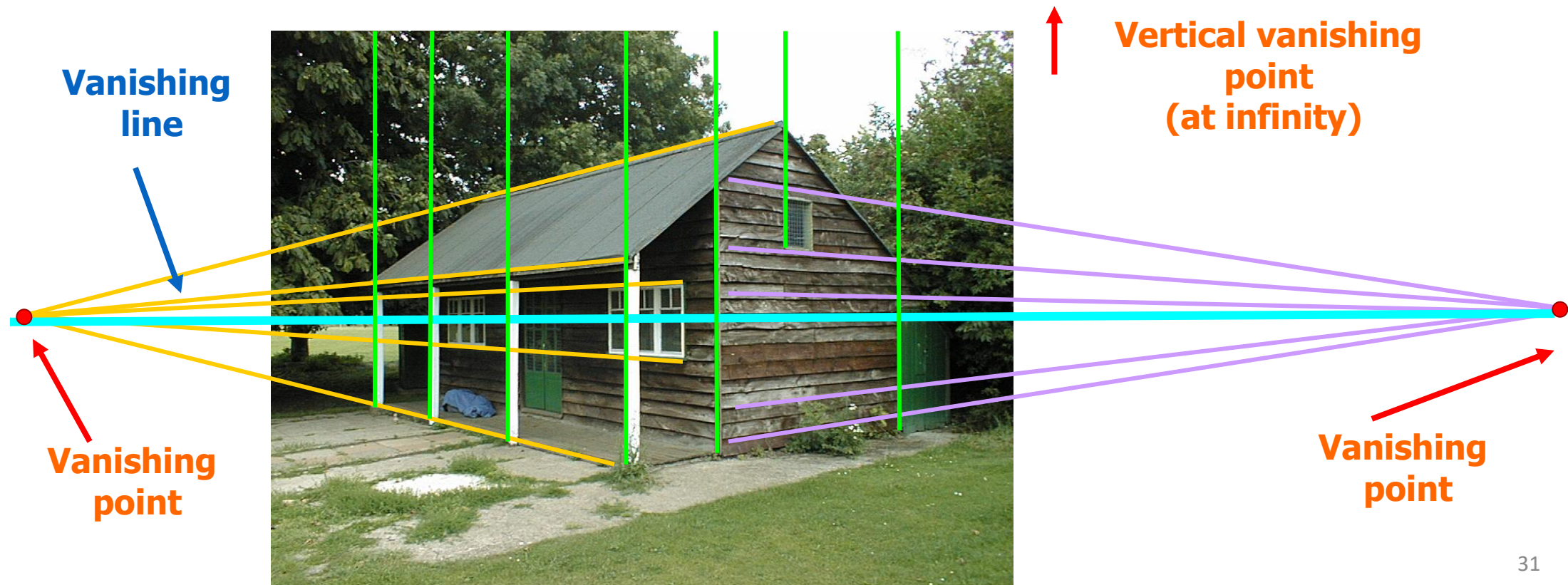


Vanishing points and lines



[Explore Vanishing Points
with interactive visualization](#)

- **Parallel lines** intersect at a “**vanishing point**” in the image
- **Parallel planes** intersect at a “**vanishing line**” in the image
- Notice that vanishing points can fall both inside or outside the image

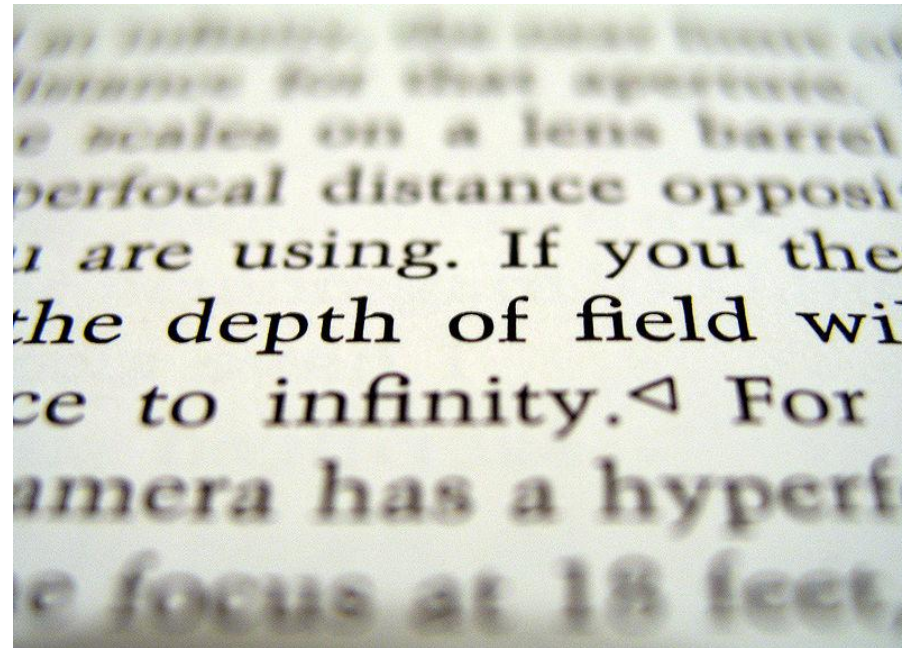
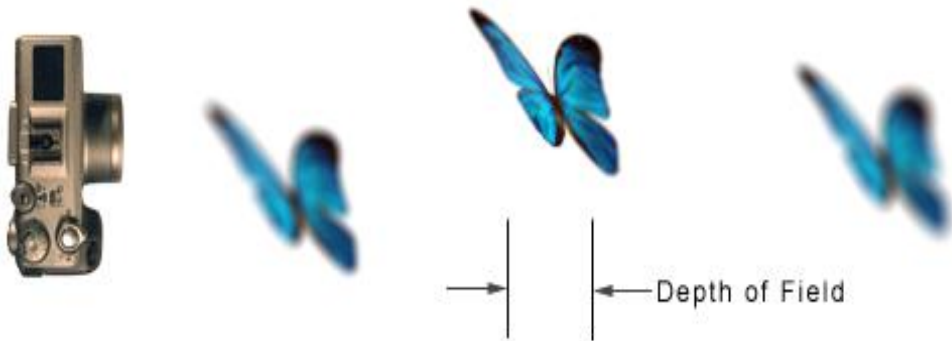


Today's Outline

- Image Formation
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Focus and Depth of Field

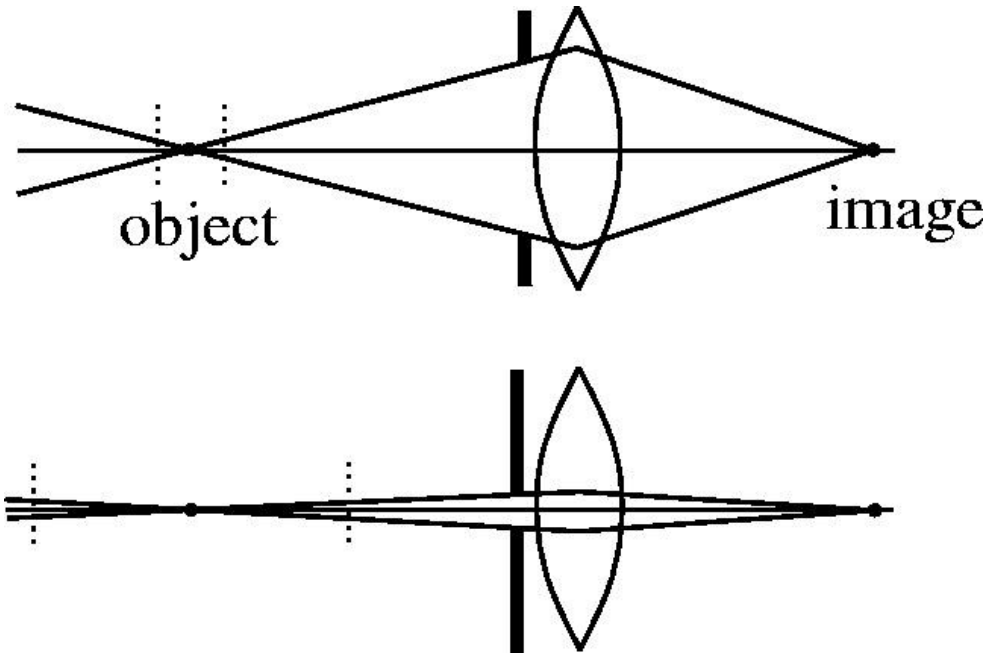
The **Depth of Field** is the distance between the **nearest and farthest objects** in a scene that appear **acceptably sharp** in an image



Depth of field

Focus and Depth of Field

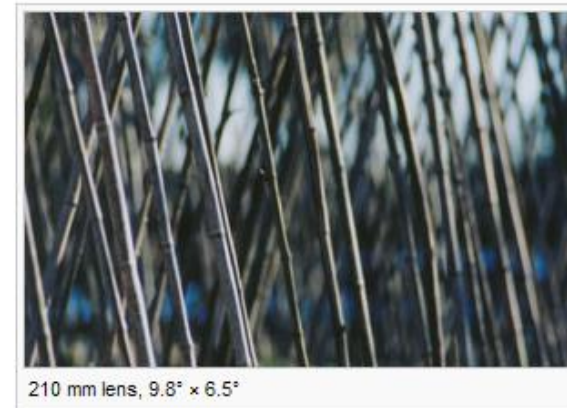
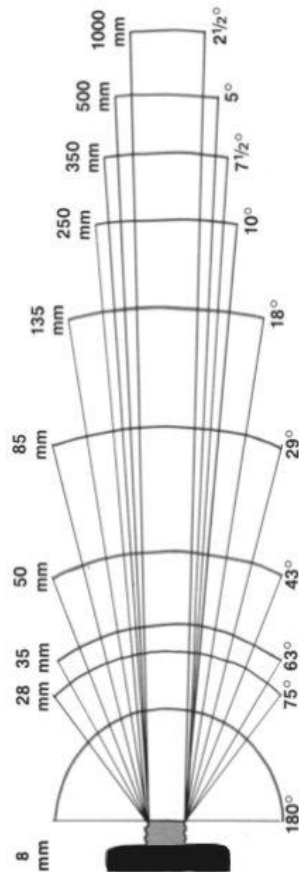
A **smaller aperture increases the depth of field** but **reduces the amount of light** into the camera: recall the definition of blur circle (it reduces with aperture)



What is the depth of field of an ideal pinhole camera?

Field of View (FOV)

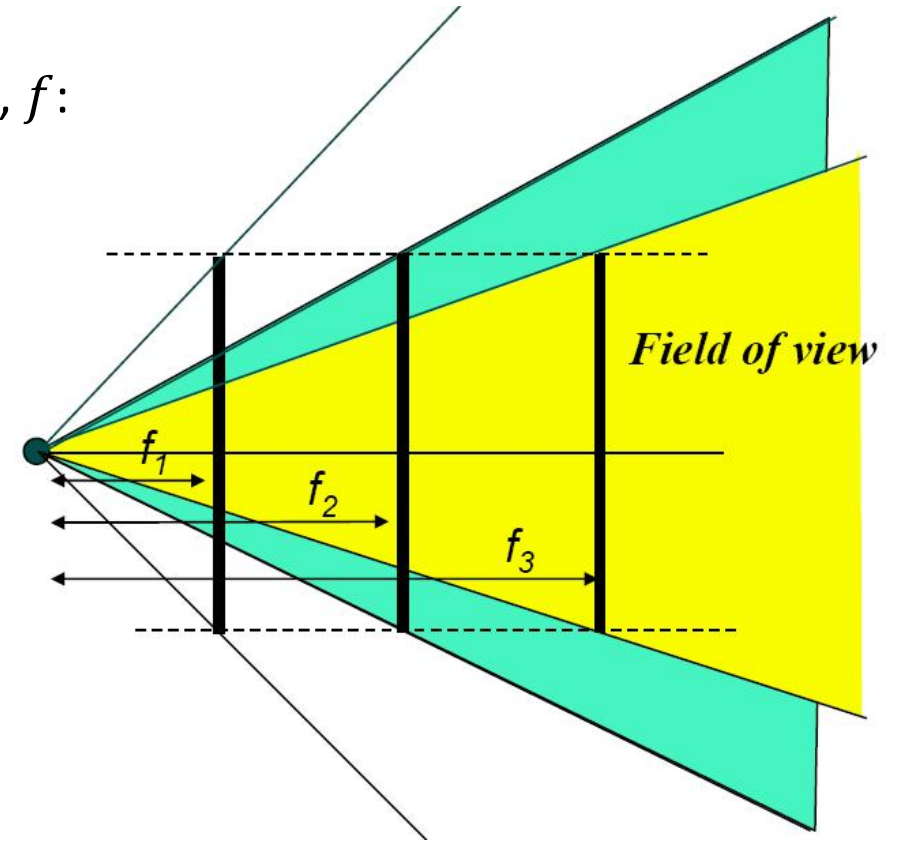
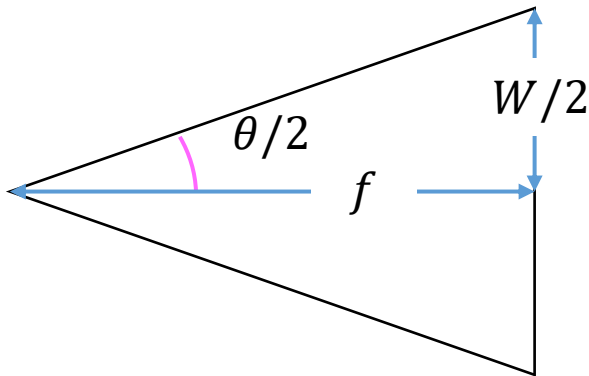
The FOV is the angular portion of 3D scene seen by the camera



Field of View (FOV)

- As the focal length f gets smaller, the FOV increases
- As the focal length f gets larger, the FOV decreases
- Relation between field of view, θ , image size, W , and focal length, f :

$$\tan \frac{\theta}{2} = \frac{W}{2f} \rightarrow f = \frac{W}{2} \left[\tan \frac{\theta}{2} \right]^{-1}$$





Curious fact: our FOV affects speed perception



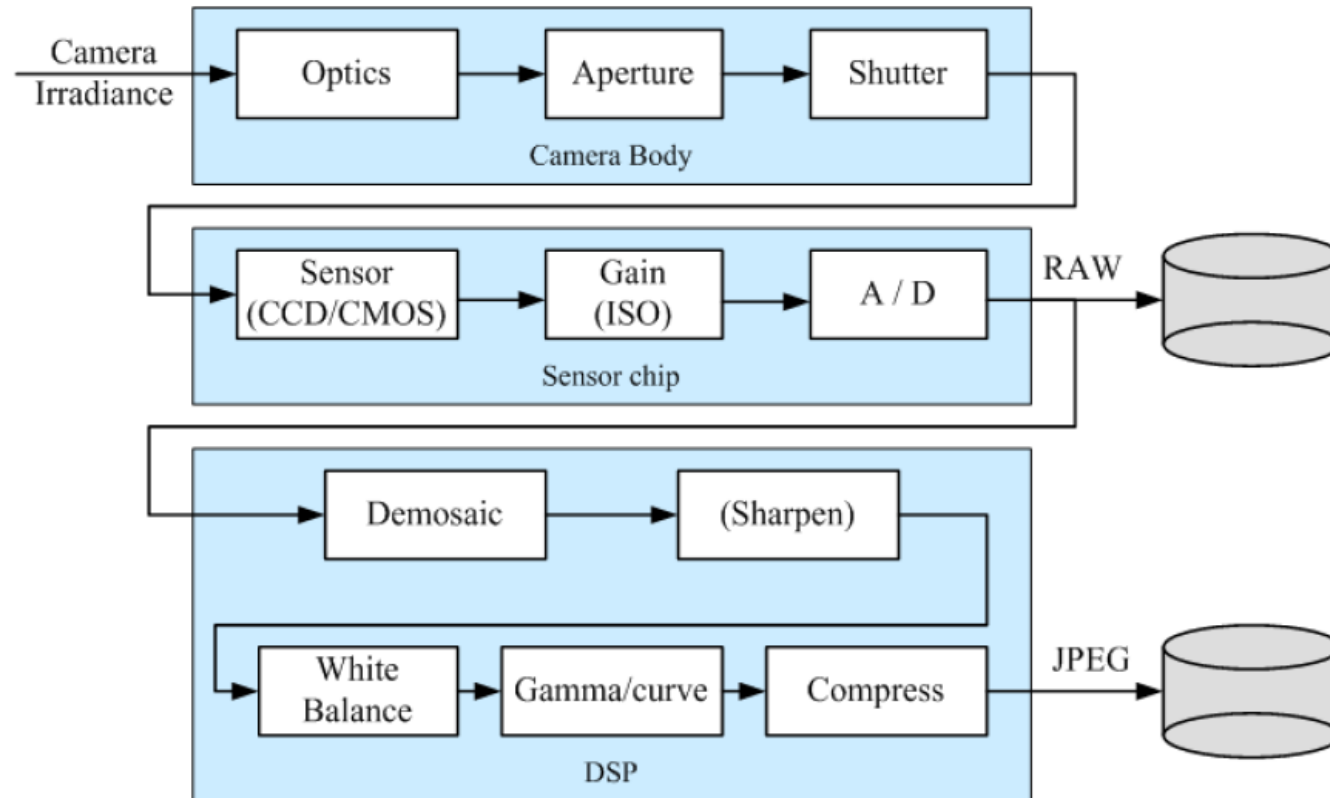
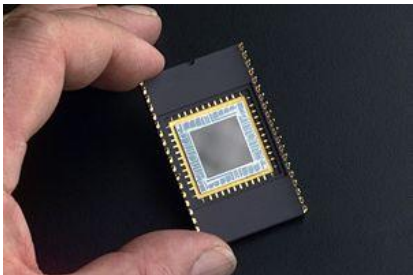
Source: <https://twitter.com/AkiyoshiKitaoka/status/1224691032173842432?s=20>

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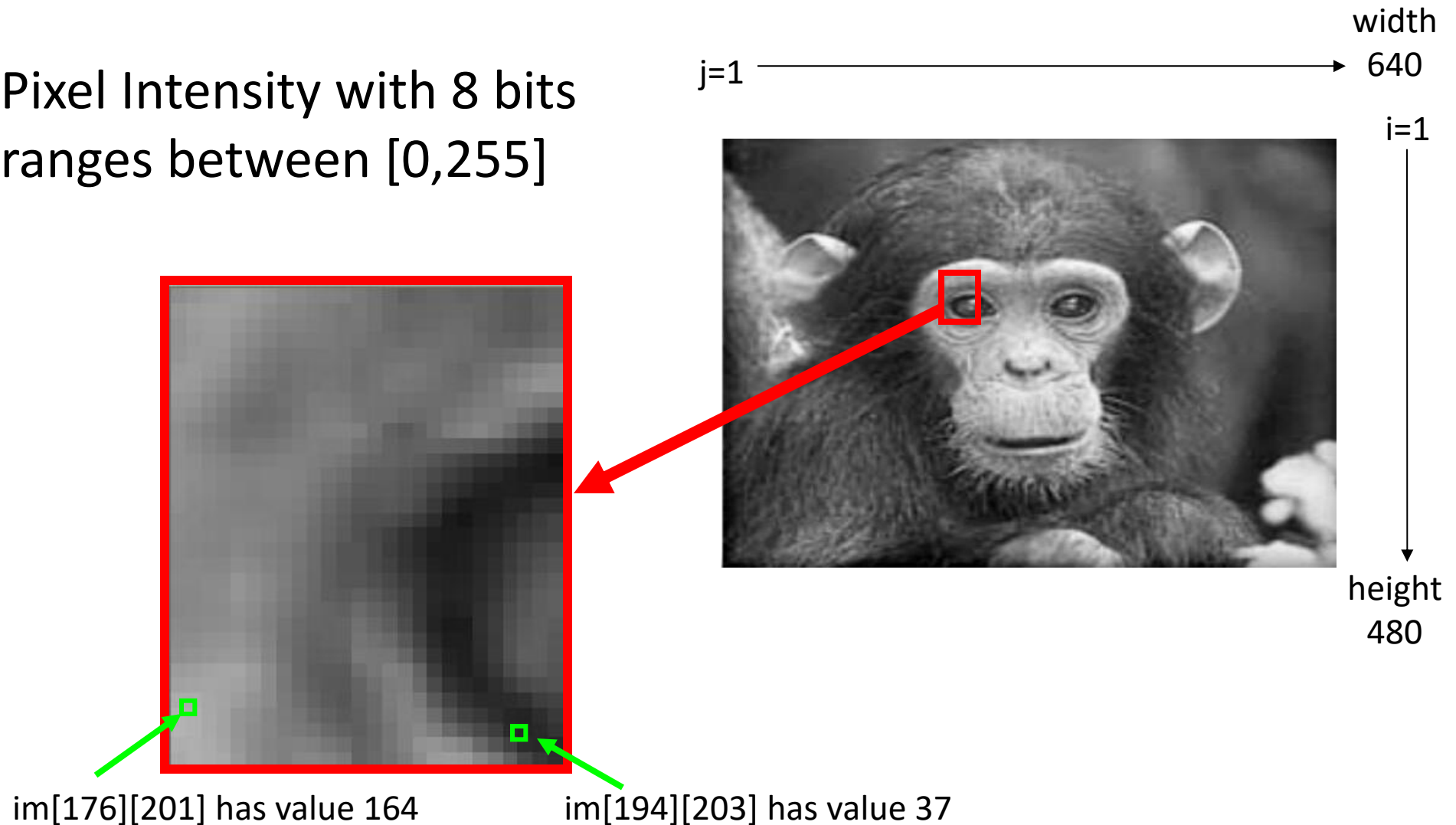
Digital camera

- In a digital camera the **film** is an array of photodiodes (CCD or CMOS) that convert photons (light energy) into electrons



Digital images

Pixel Intensity with 8 bits
ranges between [0,255]



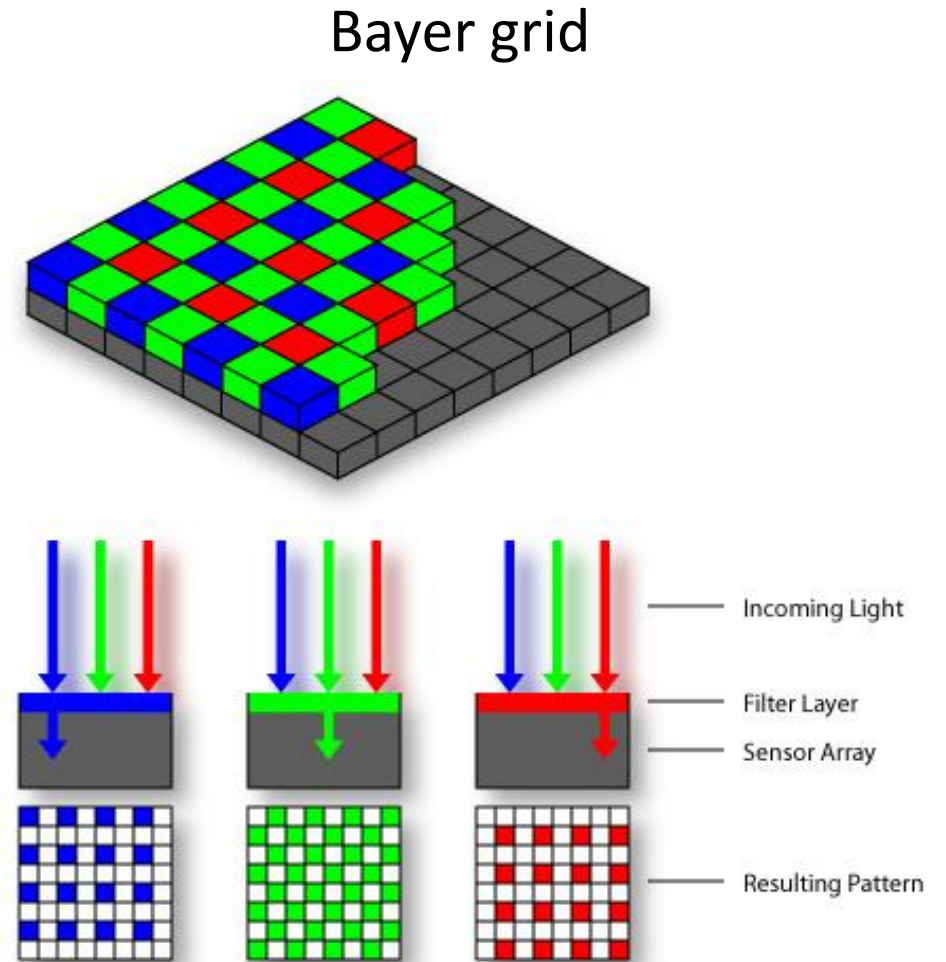
NB. Matlab and Python coordinates: [rows, cols]; OpenCV C/C++ library [cols, rows]

Color sensing in digital cameras

- The **Bayer pattern** (invented by Bayer in 1976, who worked at Kodak) places interleaved RGB filters over the pixel array
- The reason why the number of green filters is twice as that of red and blue ones is because the **luminance** signal is mostly **determined by green** values and the **human visual system** is much more **sensitive to spatial differences in luminance** than in chrominance.



For each pixel, the missing color components can be estimated from neighboring values by interpolation (**demosaicing**)



Color sensing in digital cameras

RGB color space

... but there are also many other color spaces... (e.g., YUV)



R



G



B

Rolling vs Global Shutter Camera

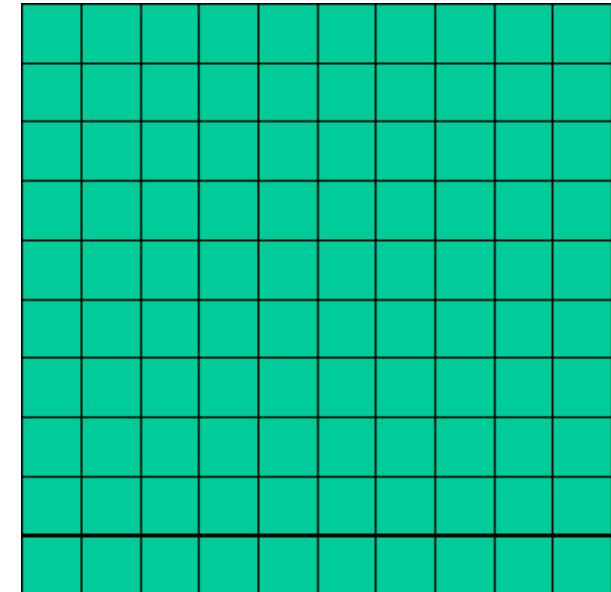
Rolling Shutter

- 👎 Rows of pixels are **exposed and read at different times**, one after the other
- 👎 May **distort (skew) moving objects**
- 👍 Cheap



Global Shutter

- 👍 All pixels are **exposed simultaneously**
- 👍 No **distortion** of moving objects
- 👎 More **costly**

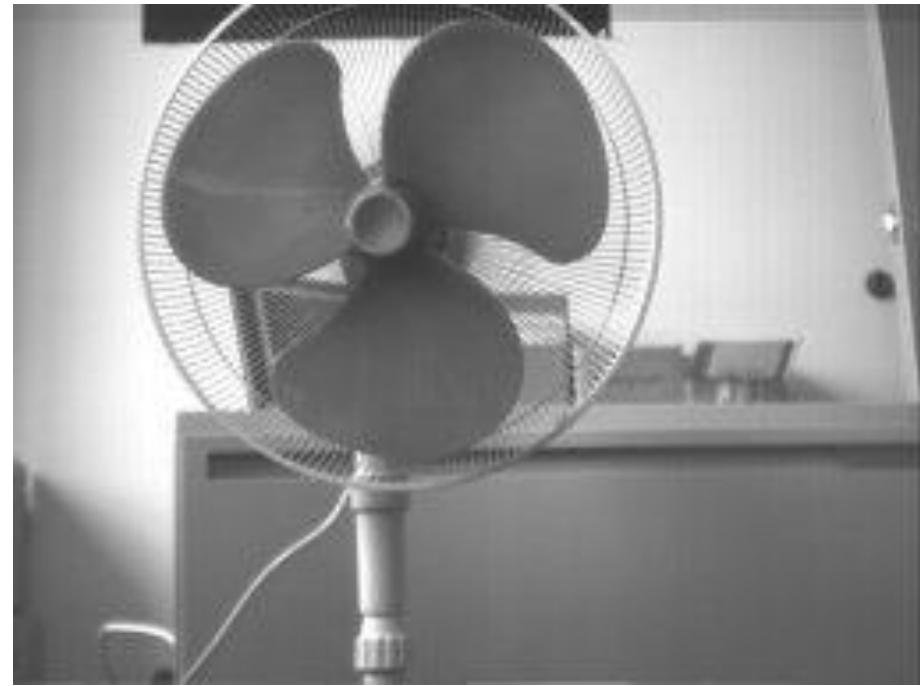


Rolling vs Global Shutter Camera

- Rolling Shutter cameras may distort (skew) moving objects
- Global Shutter cameras do not have this problem



Rolling shutter



Global shutter

Cool application of rolling shutter cameras

- High speed wave-form capturing from a guitar



Recorded from an iPhone 4 (rolling shutter):

<https://youtu.be/TKF6nFzpHBU>

An example camera datasheet

mvBlueFOX-IGC / -MLC

Technical Details



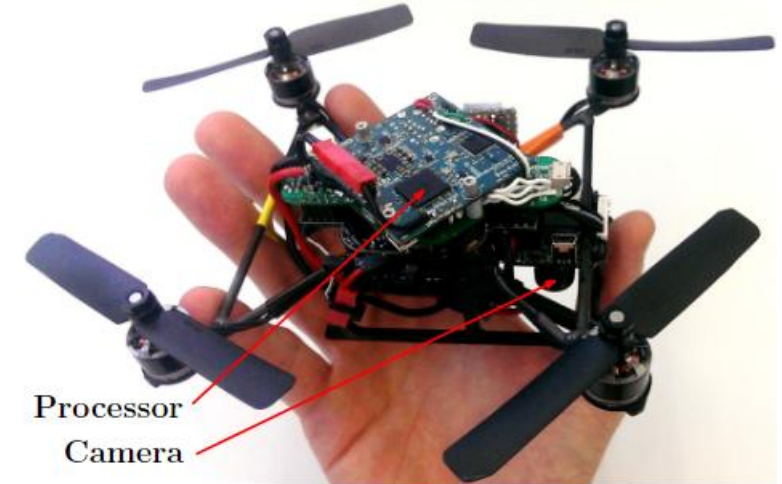
Sensors

mvBlueFOX-IGC mvBlueFOX-MLC		Resolution (H x V pixels)	Sensor size (optical)	Pixel size (μm)	Frame rate	Sensor technology	Readout type	ADC resolution / output in bits	Sensor
-200w ^{1,2}	G/C	752 x 480	1/3"	6 x 6	90	CMOS	Global	10 → 10 / 8	Aptina MT9V
-202b	G/C	1280 x 960	1/3"	3.75 x 3.75	24.6	CMOS	Global	10 → 10 / 8	Aptina MT9M
-202d ¹	G/C	1280 x 960	1/3"	3.75 x 3.75	24.6	CMOS	Rolling	10 → 10 / 8	Aptina MT9M
-205 ²	G/C	2592 x 1944	1/2.5"	2.2 x 2.2	5.8	CMOS	Global Reset	10 → 10 / 8	Aptina MT9P

¹High Dynamic Range (HDR) mode supported

²Software trigger supported

Sample: mvBlueFOX-IGC200wG means version with housing and 752 x 480 CMOS gray scale sensor.
mvBlueFOX-MLC200wG means single-board version without housing and with 752 x 480 CMOS gray scale sensor.



Nano-drone of my lab equipped
with this camera

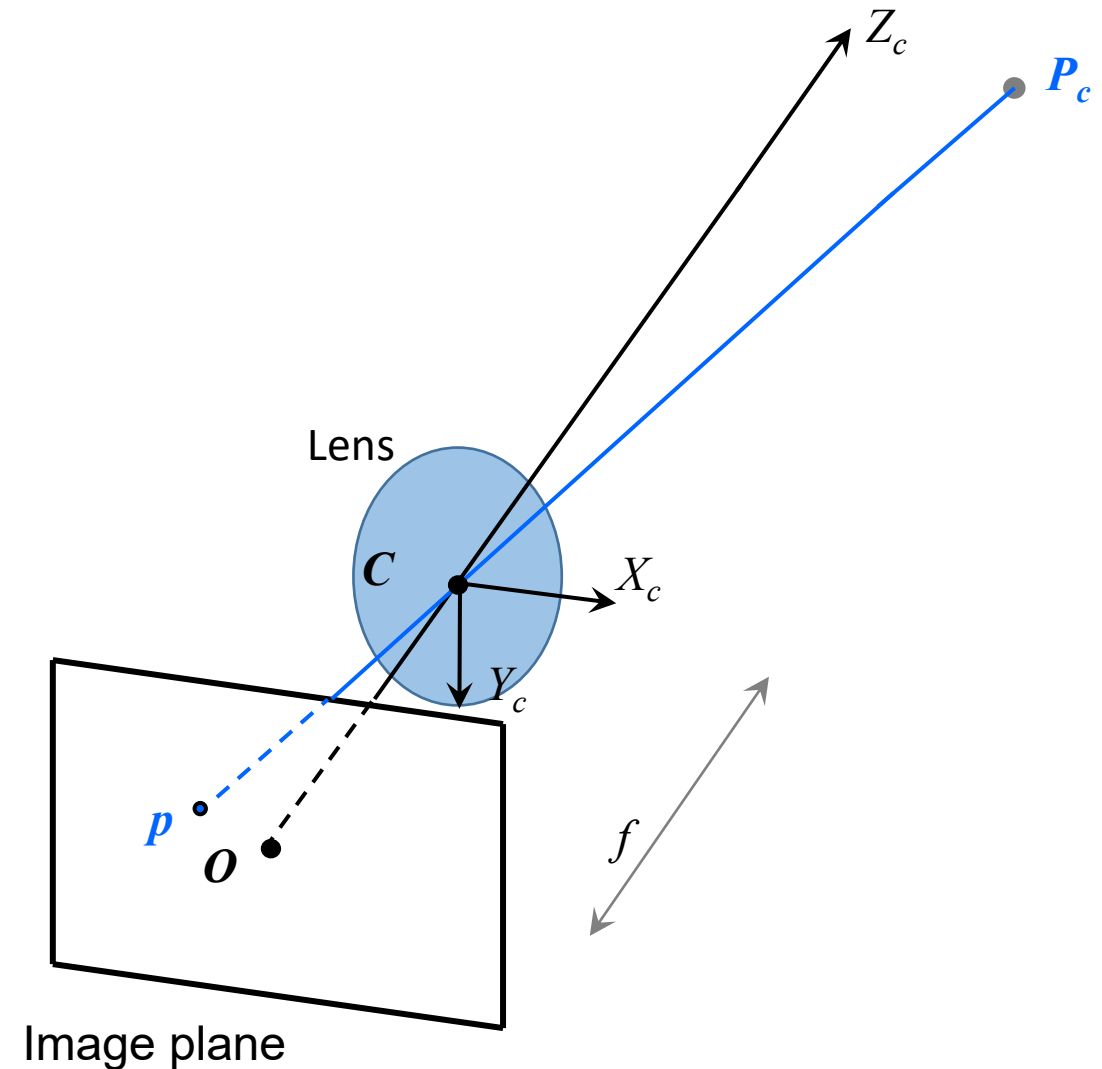
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Perspective Camera

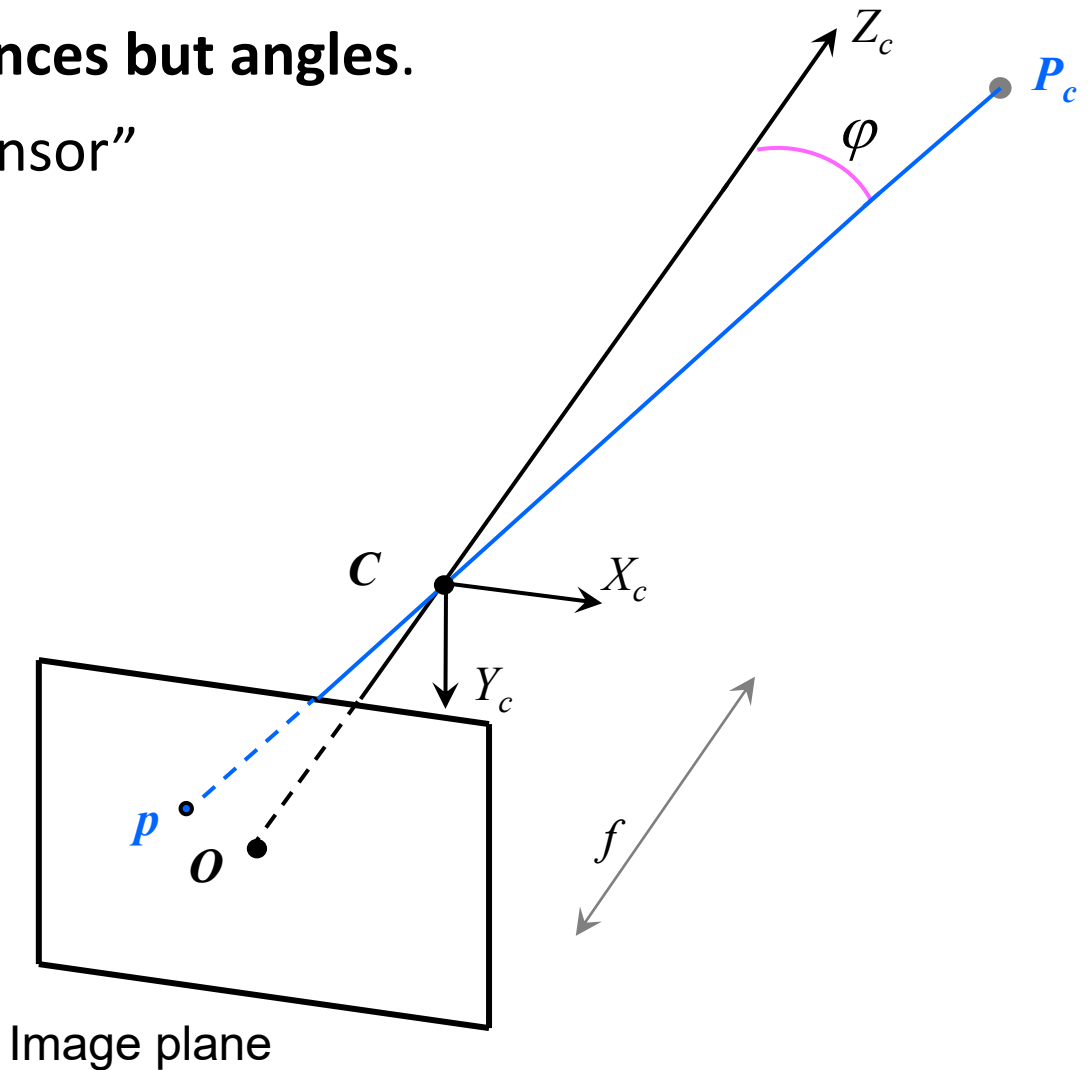
Nomenclature:

- C = optical center
= center of the lens
= center of projection
- X_c, Y_c, Z_c = axes of the camera reference frame
- Z_c = optical axis
- O = principal point
= intersection of optical axis and image plane



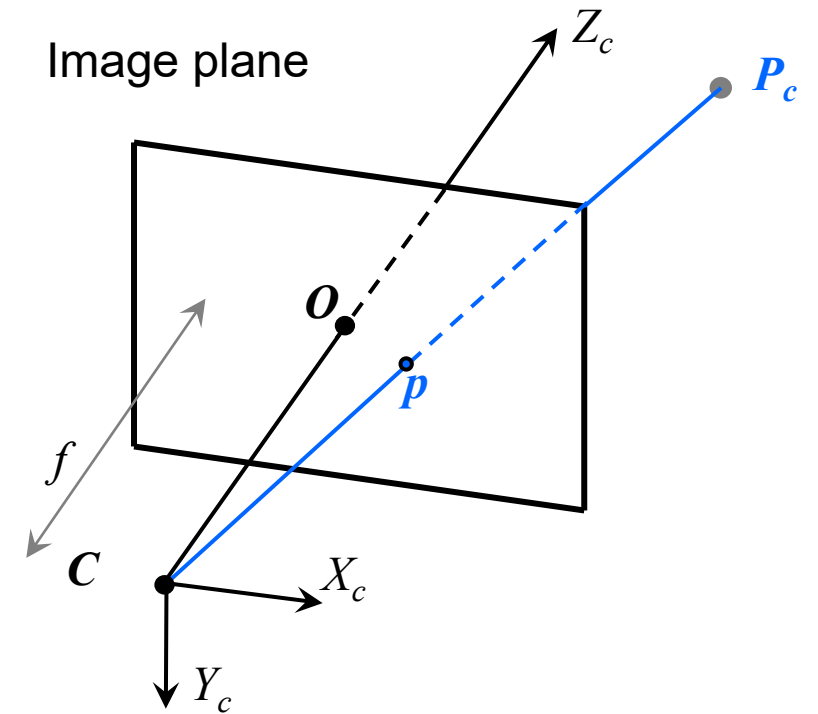
Perspective Camera

- Note that a **camera does not measure distances but angles.**
- So, we call it a “bearing sensor” or “angle sensor”



Perspective Camera

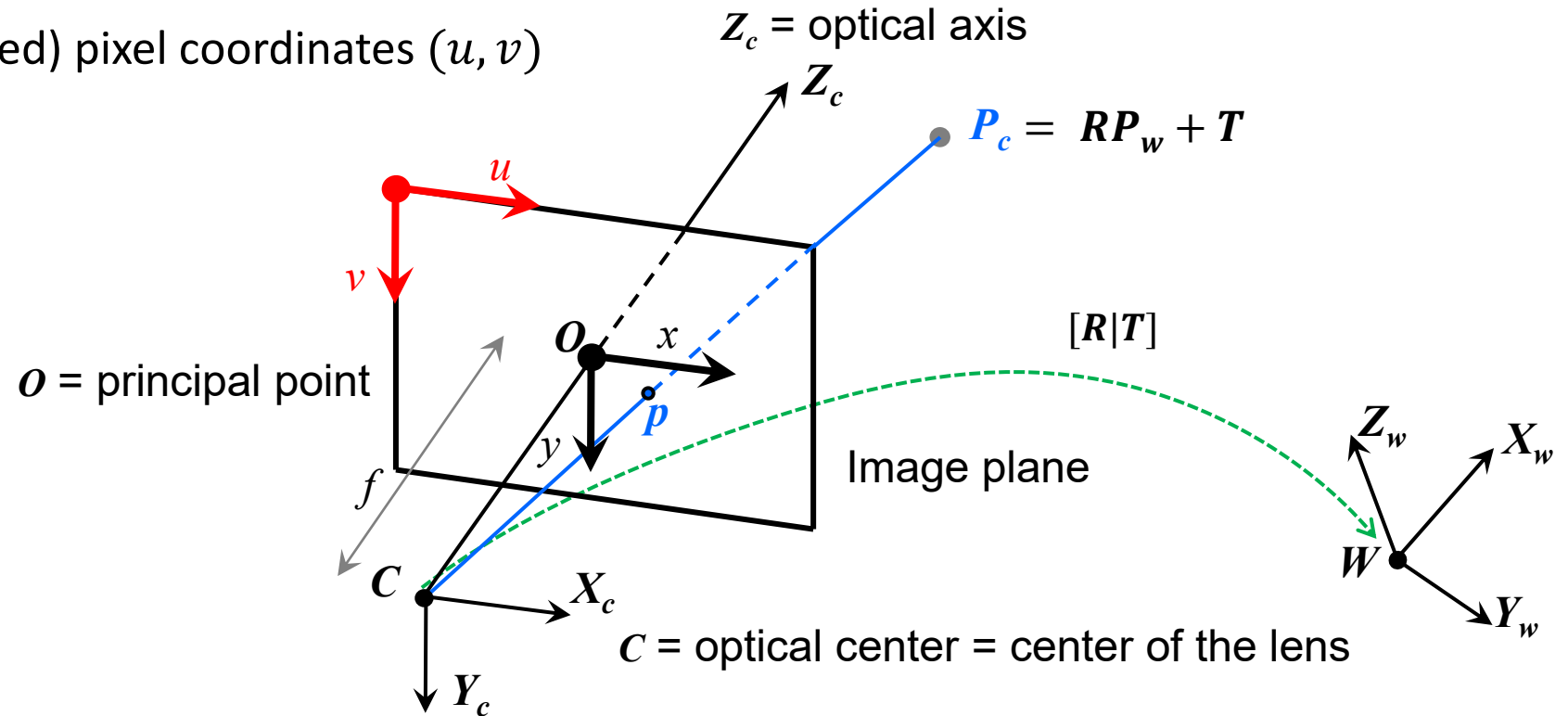
- For convenience, the **image plane** is usually **represented in front** of the lens, C , such that the image preserves the same orientation (i.e. not flipped)



From World to Pixel coordinates

Goal: Find pixel coordinates (u, v) of the projection of world point P_W onto the image plane:

- Convert world point P_W to camera point P_C through rigid body transformation $[R, T]$
- Convert P_C to image-plane coordinates (x, y)
- Convert (x, y) to (discretized) pixel coordinates (u, v)



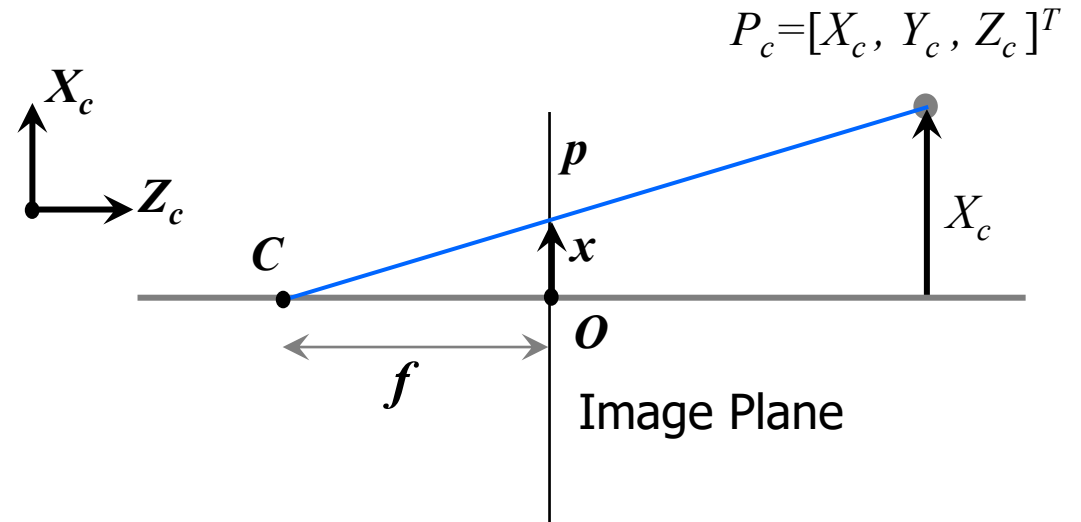
Perspective Projection – 1/4

From Camera frame to image plane coordinates (x, y)

- The Camera point $P_c = [X_c, Y_c, Z_c]^T$ projects to $p = (x, y)$ onto the image plane

- From similar triangles: $\frac{x}{f} = \frac{X_c}{Z_c} \Rightarrow x = \frac{fX_c}{Z_c}$

- Similarly, for y : $\frac{y}{f} = \frac{Y_c}{Z_c} \Rightarrow y = \frac{fY_c}{Z_c}$



Perspective Projection – 2/4

From image plane coordinates (x, y) to pixel coordinates (u, v)

- Let $O = (u_0, v_0)$ be the pixel coordinates of the camera optical center
- Let k_u, k_v be the pixel conversion factors (inverse of the pixel-size along x and y)

$$u = u_0 + k_u x \rightarrow u = u_0 + \frac{k_u f X_C}{Z_C}$$

$$v = v_0 + k_v y \rightarrow v = v_0 + \frac{k_v f Y_C}{Z_C}$$

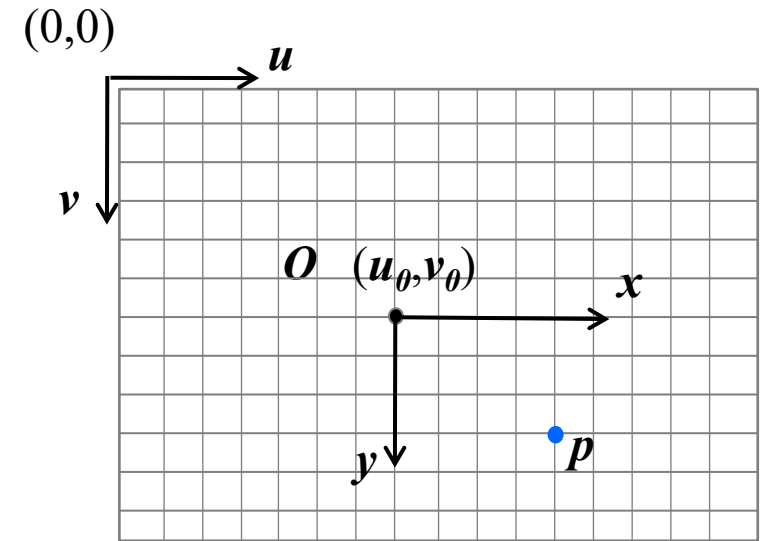


Image plane

Perspective Projection – 2/4

From image plane coordinates (x, y) to pixel coordinates (u, v)

- Let $O = (u_0, v_0)$ be the pixel coordinates of the camera optical center
- Let k_u, k_v be the pixel conversion factors (inverse of the pixel-size along x and y)

$$u = u_0 + k_u x \rightarrow u = u_0 + \frac{\alpha_u X_C}{Z_C}$$

Focal lengths
(expressed in **pixels**)

$$v = v_0 + k_v y \rightarrow v = v_0 + \frac{\alpha_v Y_C}{Z_C}$$

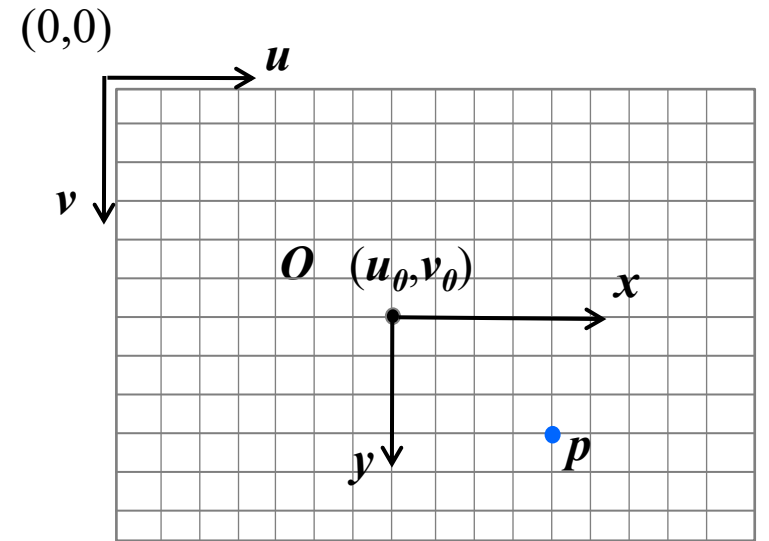


Image plane

Perspective Projection – 3/4

- Use **Homogeneous Coordinates** for linear mapping from 3D to 2D:

$$p = \begin{pmatrix} u \\ v \end{pmatrix} \Rightarrow \tilde{p} = \begin{bmatrix} \tilde{u} \\ \tilde{v} \\ \tilde{w} \end{bmatrix} = \lambda \begin{bmatrix} u \\ v \\ 1 \end{bmatrix}$$

- Using matrix form and homogeneous coordinates, the perspective projection becomes:

$$\lambda \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} \alpha_u & 0 & u_0 \\ 0 & \alpha_v & v_0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} X_c \\ Y_c \\ Z_c \end{bmatrix}$$

This matrix is called “**Calibration matrix**” or “**Intrinsic Parameter Matrix**” and is often denoted as ***K***

In the past it was common to assume a skew factor ($K_{12} \neq 0$) to account for possible skew in the pixel manufacturing process and misalignment between lens and sensor planes.

However, the camera manufacturing process today is so good that we can safely assume $K_{12} = 0$ and $\alpha_u = \alpha_v$ (i.e., square pixels).

Perspective Projection – 3/4

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$$p = \begin{pmatrix} u \\ v \end{pmatrix} \Rightarrow \tilde{p} = \begin{bmatrix} \tilde{u} \\ \tilde{v} \\ \tilde{w} \end{bmatrix} = \lambda \begin{bmatrix} u \\ v \\ 1 \end{bmatrix}$$

- Using matrix form and homogeneous coordinates, the perspective projection becomes:

$$\lambda \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = K \begin{bmatrix} X_c \\ Y_c \\ Z_c \end{bmatrix}$$

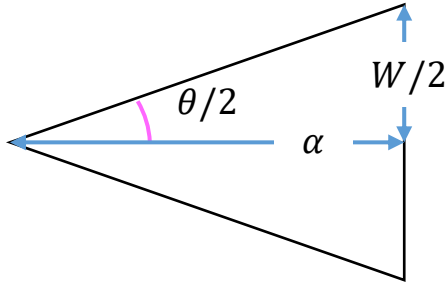
Compact form

Exercise 1

- **Determine the Intrinsic Parameter Matrix (K)** of a digital camera with an image size 640×480 pixels and a horizontal field of view of 90°
- Assume square pixels and the principal point as the center of the diagonals
- What is the vertical field of view?
- What's the projection on the image plane of $P_c = [1, 1, 2]^T$

Exercise 1 - Solution

- Recall:



$$\tan \frac{\theta}{2} = \frac{W}{2\alpha} \rightarrow \alpha = \frac{W}{2} \left[\tan \frac{\theta}{2} \right]^{-1} = \frac{640}{2 \tan \frac{\pi/2}{2}} = 320 \text{ [pixels]}$$

$\rightarrow \alpha_u = \alpha_v = \alpha = 320 \text{ pixels}$ (because pixels are square)

- Because the Principal Point is center of the diagonals $\rightarrow O = (u_0, v_0) = \left(\frac{640}{2}, \frac{480}{2} \right) = (320, 240)$

$$\rightarrow K = \begin{bmatrix} 320 & 0 & 320 \\ 0 & 320 & 240 \\ 0 & 0 & 1 \end{bmatrix}$$

- Vertical FOV: $\theta_V = 2 \tan^{-1} \frac{H}{2\alpha} = 2 \tan^{-1} \frac{480}{2 \cdot 320} = 73.74^\circ$

Exercise 1 - Solution

- What's the projection on the image plane of $P_c = [1, 1, 2]^T$

- **Solution 1** (using **Perspective Projection Equation in plane form**)

$$u = u_0 + \frac{\alpha X_C}{Z_C} = 320 + \frac{320 \cdot 1}{2} = 480 \text{ [pixels]}$$

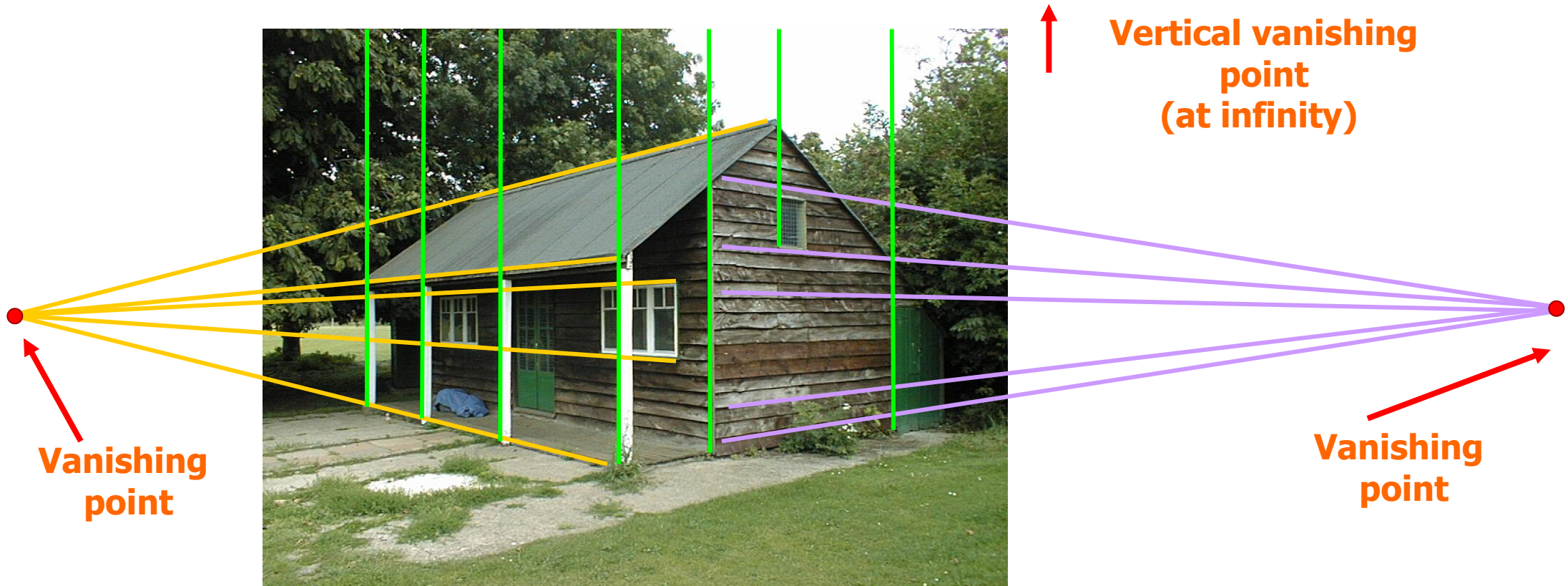
$$v = v_0 + \frac{\alpha Y_C}{Z_C} = 240 + \frac{320 \cdot 1}{2} = 400 \text{ [pixels]}$$

- **Solution 2** (using **Perspective Projection Equation in matrix form**)

$$\begin{bmatrix} \lambda u \\ \lambda v \\ \lambda \end{bmatrix} = \begin{bmatrix} 320 & 0 & 320 \\ 0 & 320 & 240 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 960 \\ 800 \\ 2 \end{bmatrix} \rightarrow \begin{aligned} u &= \frac{960}{2} = 480 \text{ [pixels]} \\ v &= \frac{800}{2} = 400 \text{ [pixels]} \end{aligned}$$

Exercise 2

- Prove that world's parallel lines intersect at a vanishing point in the camera image



Exercise 2 - Solution

- World's parallel lines with direction (l, m, n) , passing through (X_0, Y_0, Z_0) , are described by these equations:

$$\begin{aligned}X &= X_0 + sl \\Y &= Y_0 + sm \\Z &= Z_0 + sn\end{aligned}$$

- Let's consider the perspective projection equation in plane form:

$$u = u_0 + \alpha \frac{X}{Z} \quad , \quad v = v_0 + \alpha \frac{Y}{Z}$$

- Substitute line equations into the perspective projection equation and compute limit for $s \rightarrow \infty$

$$\lim_{s \rightarrow \infty} u_0 + \alpha \frac{X_0 + sl}{Z_0 + sn} = u_0 + \alpha \frac{l}{n} \quad , \quad \lim_{s \rightarrow \infty} v_0 + \alpha \frac{Y_0 + sm}{Z_0 + sn} = v_0 + \alpha \frac{m}{n}$$

- These are the image coordinates of the vanishing point. **Notice that they only depend on the line direction** (indeed, they do not depend on (X_0, Y_0, Z_0)).

Exercise 2 - Solution

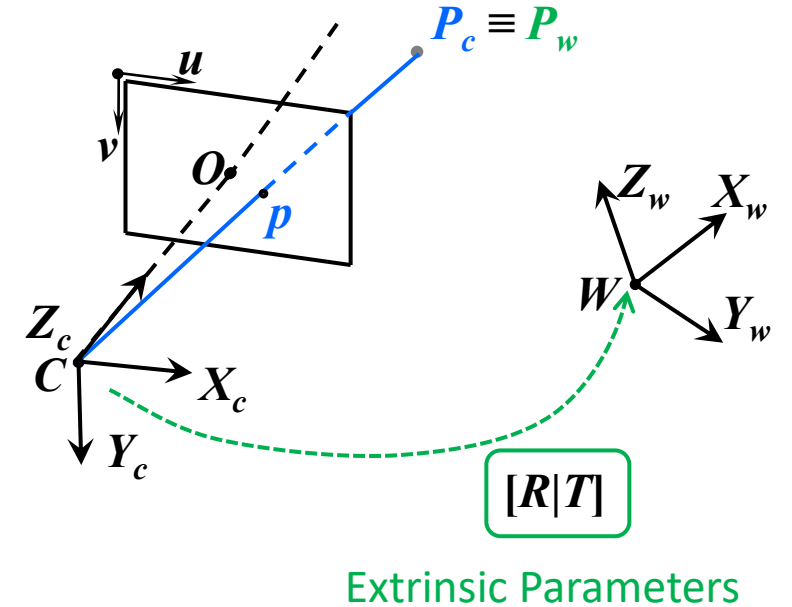
- You can verify that the solution satisfies this matrix equation. **Can you show it geometrically?**

$$\begin{bmatrix} \lambda u \\ \lambda v \\ \lambda \end{bmatrix} = \begin{bmatrix} \alpha & 0 & u_0 \\ 0 & \alpha & v_0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} l \\ m \\ n \end{bmatrix}$$

Perspective Projection – 4/4

From the World frame to the Camera frame

$$\begin{bmatrix} X_c \\ Y_c \\ Z_c \end{bmatrix} = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix} \begin{bmatrix} X_w \\ Y_w \\ Z_w \end{bmatrix} + \begin{bmatrix} t_1 \\ t_2 \\ t_3 \end{bmatrix}$$



Perspective Projection – 4/4

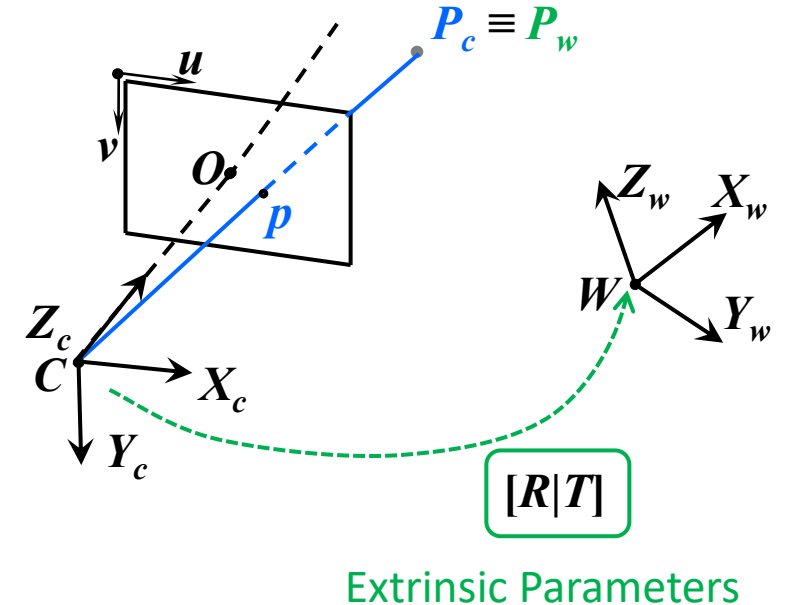
From the World frame to the Camera frame

$$\begin{bmatrix} X_c \\ Y_c \\ Z_c \end{bmatrix} = \begin{bmatrix} r_{11} & r_{12} & r_{13} & t_1 \\ r_{21} & r_{22} & r_{23} & t_2 \\ r_{31} & r_{32} & r_{33} & t_3 \end{bmatrix} \cdot \begin{bmatrix} X_w \\ Y_w \\ Z_w \\ 1 \end{bmatrix} = \left[\begin{array}{c|c} R & T \end{array} \right] \cdot \begin{bmatrix} X_w \\ Y_w \\ Z_w \\ 1 \end{bmatrix}$$

By substituting the above one into the **Perspective Projection Equation**:

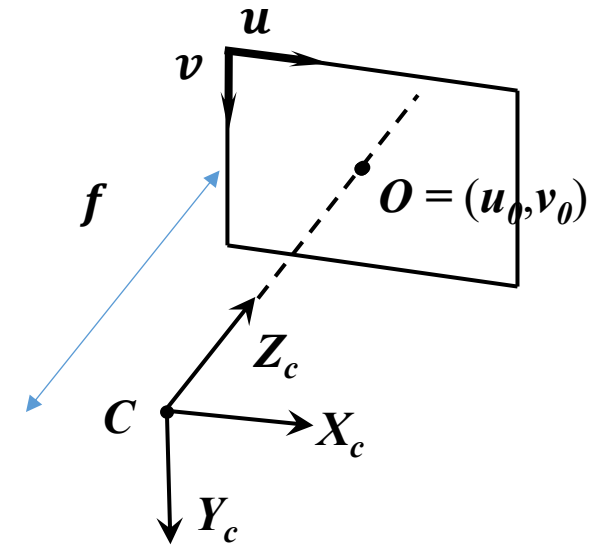
$$\lambda \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = K \begin{bmatrix} X_c \\ Y_c \\ Z_c \end{bmatrix}$$

$$\lambda \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \underbrace{K \begin{bmatrix} R & T \end{bmatrix}}_{\text{Projection Matrix (M)}} \cdot \begin{bmatrix} X_w \\ Y_w \\ Z_w \\ 1 \end{bmatrix}$$

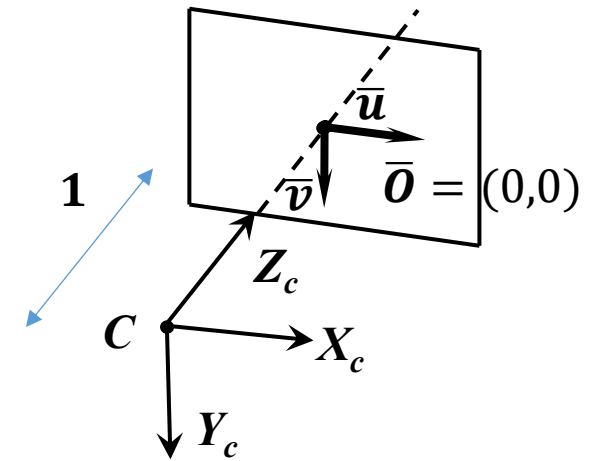


Perspective Projection Equation

- It is often convenient to use normalized image coordinates onto a **virtual image plane with focal length equal to 1 unit** (adimensional) **and origin of the pixel coordinates at the principal point**.



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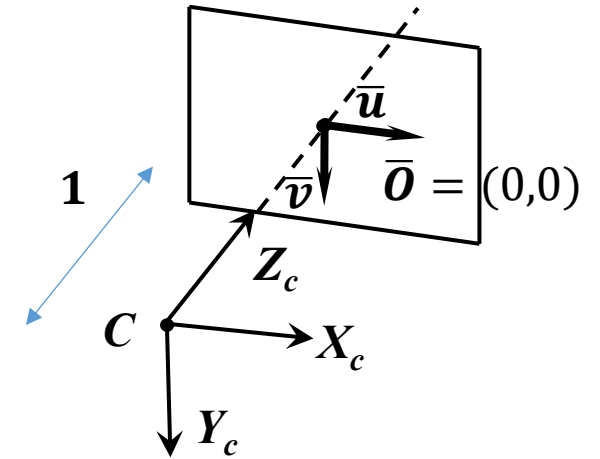
Normalized image coordinates

- It is often convenient to use normalized image coordinates onto a **virtual image plane with focal length equal to 1 unit** (adimensional) **and origin of the pixel coordinates at the principal point**.
- To do this, we just pre-multiply both terms of the perspective projection equation in camera frame coordinates by K^{-1} :

$$\lambda \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = K \begin{bmatrix} X_c \\ Y_c \\ Z_c \end{bmatrix} \Rightarrow \lambda K^{-1} \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = K^{-1} \cdot K \begin{bmatrix} X_c \\ Y_c \\ Z_c \end{bmatrix} \Rightarrow \lambda K^{-1} \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} X_c \\ Y_c \\ Z_c \end{bmatrix}$$

- We define the **unit-plane normalized image coordinates** $(\bar{u}, \bar{v})$:

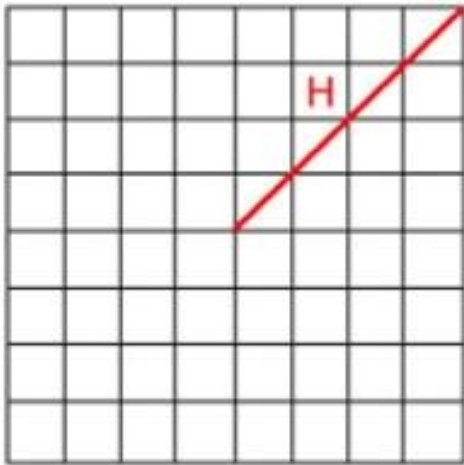
$$\begin{bmatrix} \bar{u} \\ \bar{v} \\ 1 \end{bmatrix} = K^{-1} \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} \frac{1}{\alpha} & 0 & -\frac{u_0}{\alpha} \\ 0 & \frac{1}{\alpha} & -\frac{v_0}{\alpha} \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} \frac{u-u_0}{\alpha} \\ \frac{v-v_0}{\alpha} \\ 1 \end{bmatrix} \Rightarrow \lambda \begin{bmatrix} \bar{u} \\ \bar{v} \\ 1 \end{bmatrix} = \begin{bmatrix} X_c \\ Y_c \\ Z_c \end{bmatrix}$$



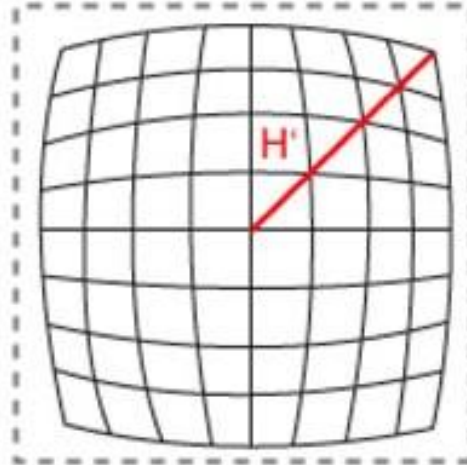
Today's Outline

- Image Formation
- Other camera parameters
- Digital camera
- Perspective camera model
- Lens distortion

Radial Distortion

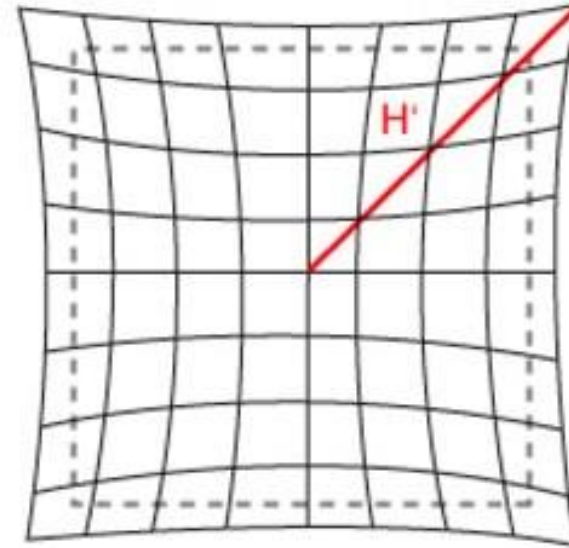


No distortion



Negative radial distortion
(Barrel distortion)

$$k_1 < 0$$



Positive radial distortion
(Pincushion distortion)

$$k_1 > 0$$

Radial Distortion in the OpenCV and Matlab Camera Models

- The standard model of radial distortion is a transformation **from the ideal (non-distorted)** coordinates (u, v) **to the real (distorted)** coordinates (u_d, v_d)
- For a given non distorted image point (u, v) , the amount of distortion is a nonlinear function of its distance r from the principal point. For most lenses, **this simple quadratic model of radial distortion is sufficient:**

$$\begin{bmatrix} u_d \\ v_d \end{bmatrix} = (1 + k_1 r^2) \begin{bmatrix} u - u_0 \\ v - v_0 \end{bmatrix} + \begin{bmatrix} u_0 \\ v_0 \end{bmatrix}$$

where

$$r^2 = (u - u_0)^2 + (v - v_0)^2$$

Radial & Tangential Distortion in the OpenCV and Matlab Camera Models

- **Radial Distortion:** Depending on the amount of distortion (and thus on the camera field of view), higher order terms can be introduced for the radial distortion
- **Tangential Distortion:** if the lens is misaligned (not perfectly parallel to the image sensor), a *non-radial* (tangential) distortion is introduced

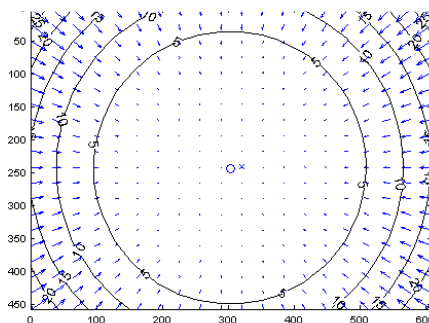
Radial distortion

Tangential distortion

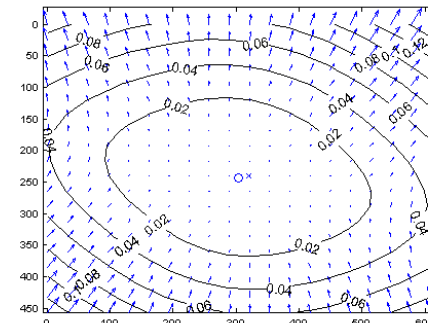
$$\begin{bmatrix} u_d \\ v_d \end{bmatrix} = (1 + k_1 r^2 + k_2 r^4 + k_3 r^6) \begin{bmatrix} u - u_0 \\ v - v_0 \end{bmatrix} + \begin{bmatrix} 2k_4(u - u_0)(v - v_0) + k_5(r^2 + 2(u - u_0)^2) \\ k_4(r^2 + 2(v - v_0)^2 + 2k_5(u - u_0)(v - v_0) \end{bmatrix} + \begin{bmatrix} u_0 \\ v_0 \end{bmatrix}$$

This formula
won't be asked at
the exam 😊

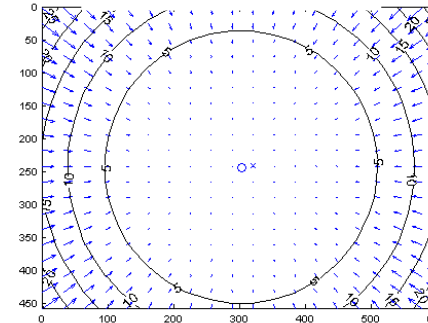
Complete distortion model



Tangential component



Radial component



The **left figure** shows the impact of the **complete distortion model (radial + tangential)** on each pixel of the image. Each arrow represents the effective displacement of a pixel induced by the lens distortion. Observe that points at the corners of the image are displaced by as much as **25 pixels**. The **center figure** shows the impact of the **tangential component of distortion**. On this plot, the **maximum induced displacement is 0.14 pixel** (at the upper left corner of the image). Finally, the **right figure** shows the impact of the **radial component of distortion**. This plot is very similar to the full distortion plot, showing that the tangential component could very well be discarded in the complete distortion model. On the three figures, the **cross** indicates the **center of the image** (i.e., the center of the two diagonals), and the **circle** the location of the **principal point**.

Summary: Perspective projection equations

- A world's 3D point $P_w = (X_w, Y_w, Z_w)$ projects into the image point $p = (u, v)$

$$\lambda \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = K [R | T] \cdot \begin{bmatrix} X_w \\ Y_w \\ Z_w \\ 1 \end{bmatrix} \quad \text{where} \quad K = \begin{bmatrix} \alpha & 0 & u_0 \\ 0 & \alpha & v_0 \\ 0 & 0 & 1 \end{bmatrix}$$

- To account for radial distortion, distorted pixel coordinates (u_d, v_d) can be obtained as:

$$\begin{bmatrix} u_d \\ v_d \end{bmatrix} = (1 + k_1 r^2) \begin{bmatrix} u - u_0 \\ v - v_0 \end{bmatrix} + \begin{bmatrix} u_0 \\ v_0 \end{bmatrix} \quad \text{where} \quad r^2 = (u - u_0)^2 + (v - v_0)^2$$

- For convenience, the projection of P_w into pixel coordinates, including lens distortion, is usually denoted:

$$\begin{bmatrix} u_d \\ v_d \end{bmatrix} = \pi(P_w, K, k_1, R, T)$$

Application of Image Formation in Film Making: *Virtual Production using LED Volumes*



The Mandalorian series

Application of Image Formation in Film Making:

Virtual Production using LED Volumes

- Instead of shooting actors in front of a green screen, huge high-resolution LED walls (and sometimes ceilings) display photorealistic, real-time rendered environments created in **Unreal Engine**.
- It was developed by **Industrial Light & Magic (ILM)**.
- Key aspects:
 - The LED screens show the **background in correct perspective, so the camera sees a realistic world that moves naturally with parallax**.
 - Real light from the screens illuminates actors and props, so the lighting is perfectly integrated without heavy postproduction.
 - Directors can adjust environments on the fly (time of day, weather, scenery).
 - This approach revolutionized VFX because it merges **live-action filmmaking** with **real-time game engine rendering**, cutting down on costly location shoots and heavy green-screen compositing.
- Real-time rendering system
 - The core principle is the full **camera perspective projection**
 - The **camera motion on set is tracked with sub-millimeter precision** using a **motion capture system** (e.g., optical markers or inside-out tracking (V-SLAM)).
 - Its pose is fed into Unreal Engine, which re-renders the virtual 3D background from that exact viewpoint using the projection model above.

Summary (things to remember)

- This-lens equation
- From the thin lens to the pinhole camera
- Perspective Projection Equation
- Vanishing points and lines
- Intrinsic and extrinsic parameters ($\mathbf{K}, \mathbf{R}, \mathbf{T}$)
- Homogeneous coordinates
- Normalized image coordinates
- Radial distortion

Readings

- Chapter 4 of Autonomous Mobile Robots book: [link](#)
- Ch. 2.1 and 2.3 of Szeliski book, 2nd Edition

Understanding Check

Are you able to:

- Explain what a blur circle is?
- Derive the thin lens equation and perform the pinhole approximation?
- Explain how to build an Ames room?
- Derive a relation between the field of view and the focal length?
- Proof the perspective projection equation, including lens distortion and world-to-camera projection?
- Explain normalized image coordinates and their geometric explanation?
- Define vanishing points and lines?
- Prove that parallel lines intersect at vanishing points?
- How is this used in modern filmmaking?